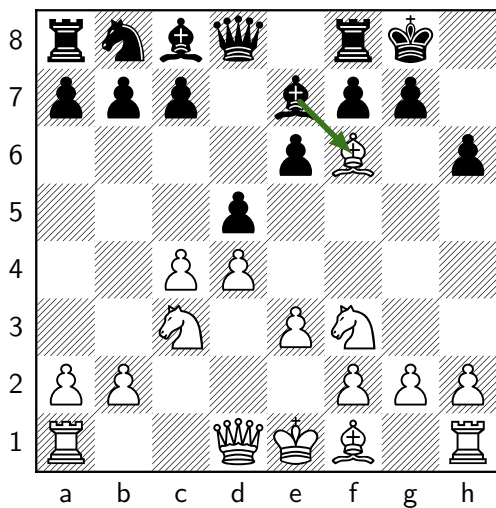


Melanie-ANALOGY — QGD, Orthodox

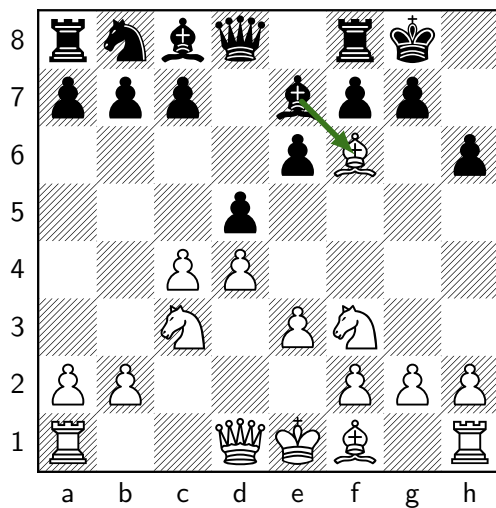
Melanie (Deep_Vision)

Abstract

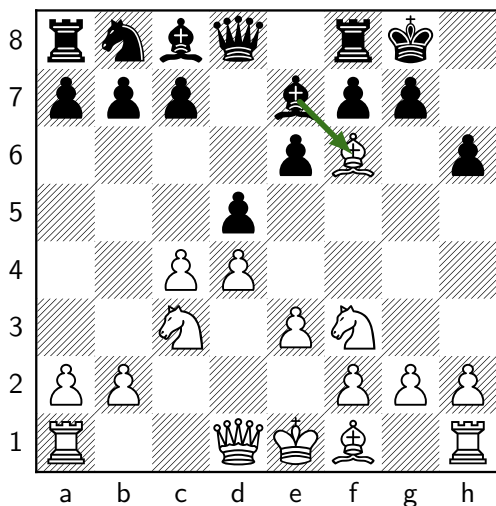
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



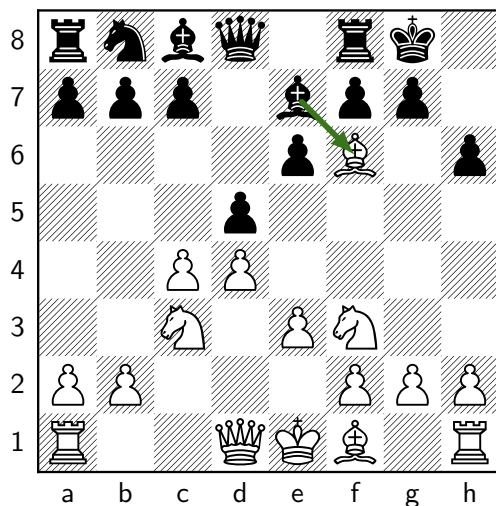
(a)



(b)

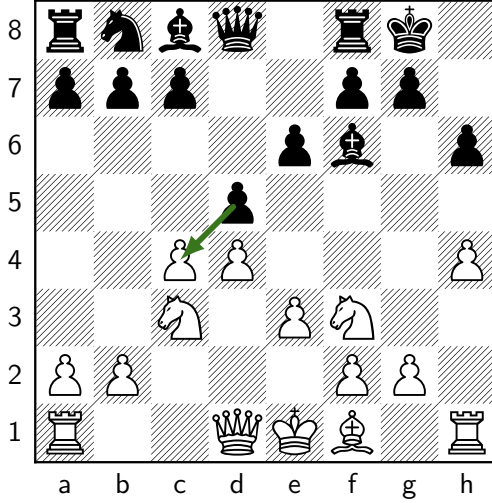


(c)

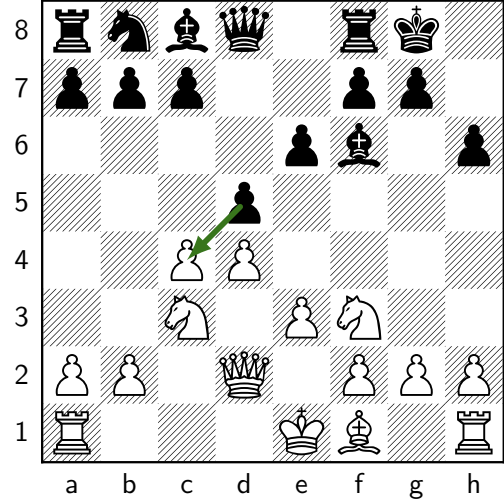


(d)

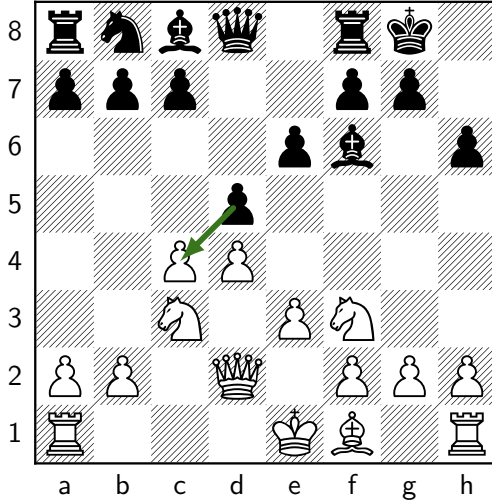
Figure 1: **Move 7, Black: Bxf6 (e7f6).** (a) Our position; Melanie plays Bxf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1759` (ply 13) [structural distance $d=0$, QAP obj=192]: the master played Bxf6; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Pa7* as their *Nb8*, our *Nb8* as their *Pa7*; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1760` (ply 13) [structural distance $d=0$, QAP obj=192]: the master played Bxf6; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Pa7* as their *Nb8*, our *Nb8* as their *Pa7*; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#815` (ply 13) [structural distance $d=0$, QAP obj=192]: the master played Bxf6; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Pa7* as their *Nb8*, our *Nb8* as their *Pa7*; transferred move e7f6 (**endorses this move**). Synthesis: 60 retrieved analogs transferred to Bxf6 (summed QAP objective 10491, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



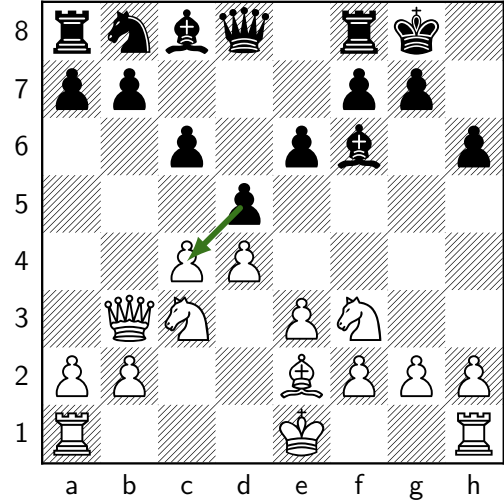
(a)



(b)

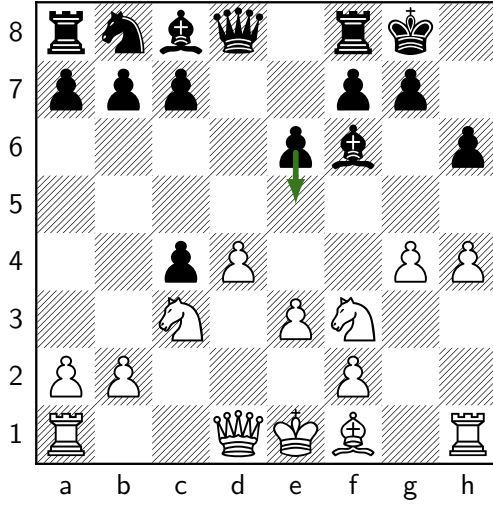


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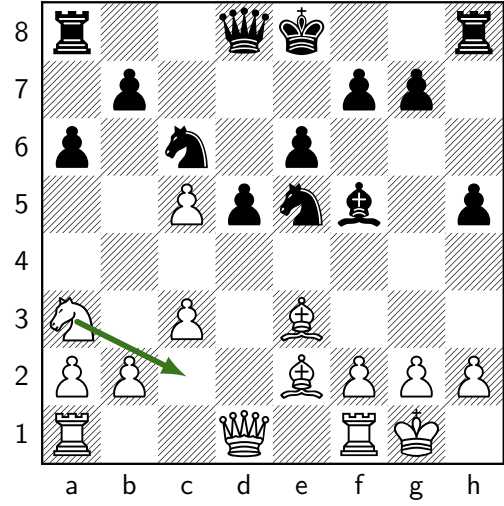


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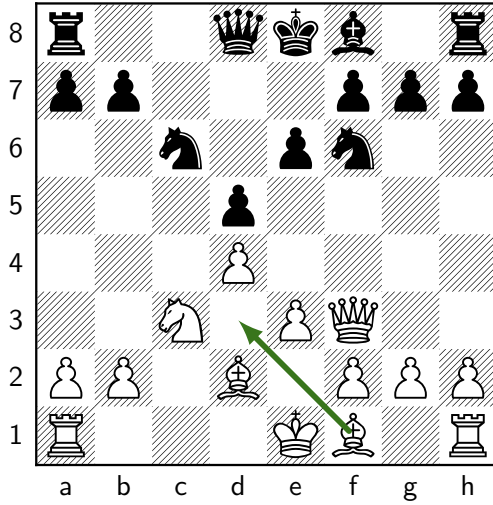
Figure 2: **Move 8, Black: dxc4.** (a) Our position; Melanie plays dxc4 (d5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1759` (ply 15) [structural distance $d=37$, QAP obj=155]: the master played dxc4; under μ our $Pd4$ plays the role of that game’s $Pd4$; transferred move d5c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1760` (ply 15) [structural distance $d=37$, QAP obj=155]: the master played dxc4; under μ our $Pd4$ plays the role of that game’s $Pd4$; transferred move d5c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2623` (ply 17) [structural distance $d=56$, QAP obj=129]: the master played dxc4; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Pf7$ as their $Ra8$, our $Pb2$ as their $Nb8$; transferred move d5c4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to dxc4 (summed QAP objective 915, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



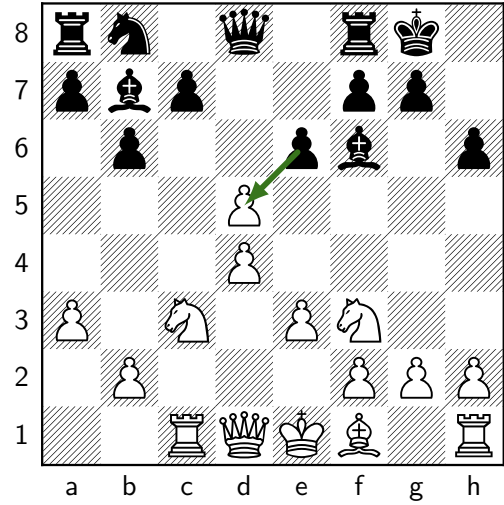
(a)



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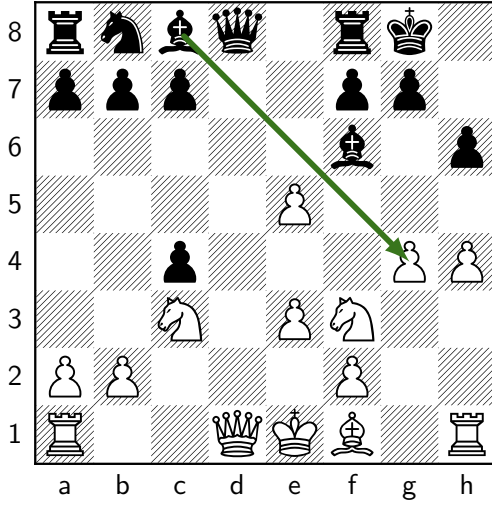


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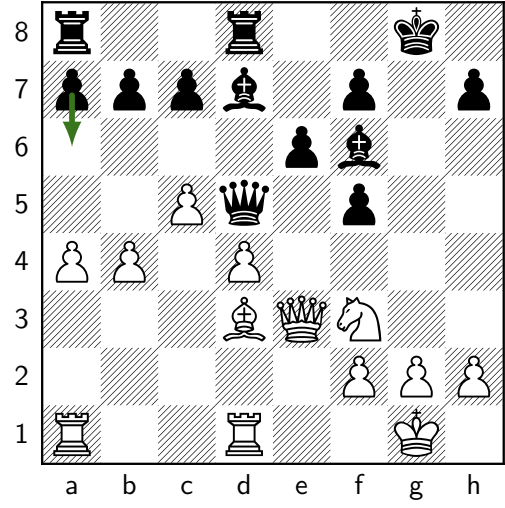


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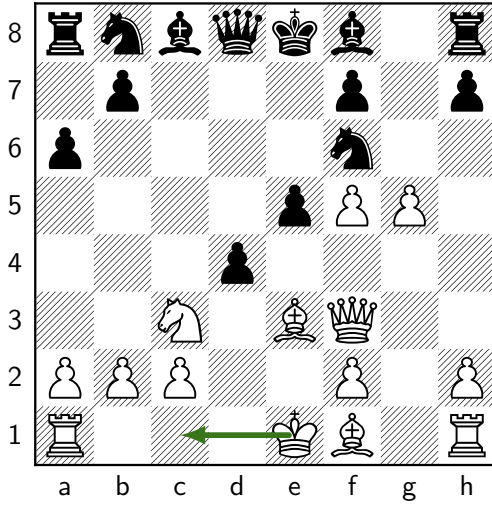
Figure 3: **Move 9, Black: e5**. (a) Our position; Melanie plays e5 (e6e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3137` (ply 26) [structural distance $d=66$, QAP obj=68]: the master played Nc2; under μ our *Pe3* plays the role of that game’s *Na3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *Ra8* as their *Pa2*; transferred move e6e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2353` (ply 16) [structural distance $d=56$, QAP obj=67]: the master played Bd3; under μ our *Pe3* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra1*, our *Ph6* as their *Rh1*; transferred move e6e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#666` (ply 19) [structural distance $d=42$, QAP obj=107]: the master played exd5; under μ our *Bf3* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pc7* as their *Nb8*; transferred move f6e7. Synthesis: 2 retrieved analogs transferred to e5 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



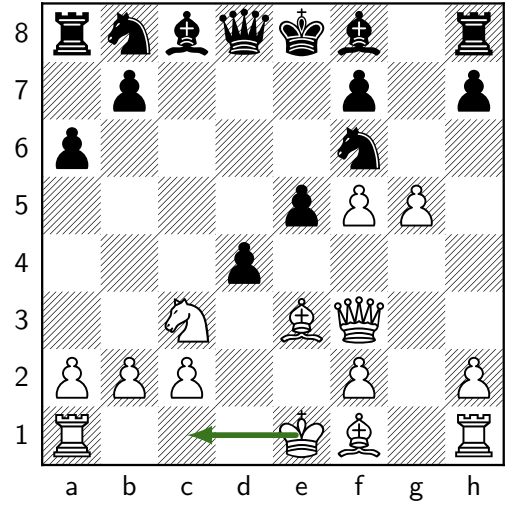
(a)



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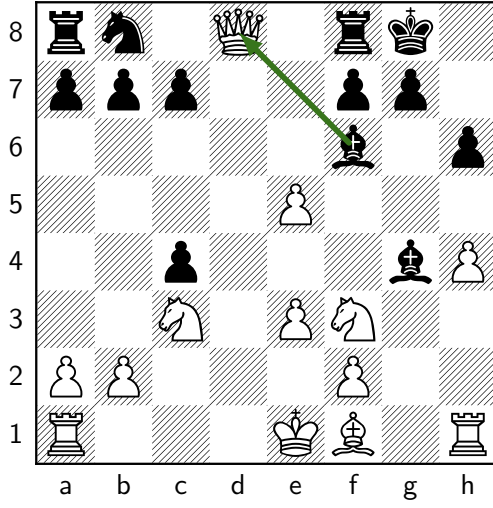


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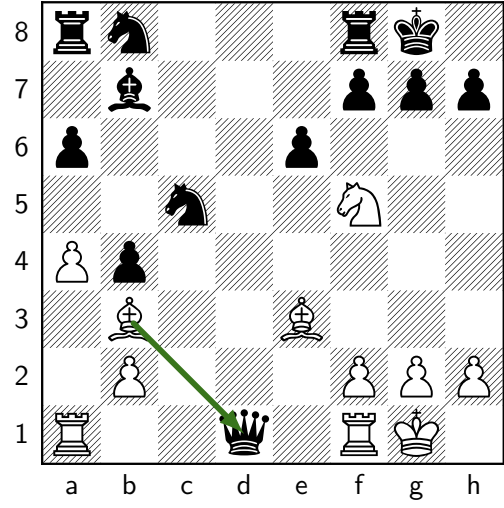


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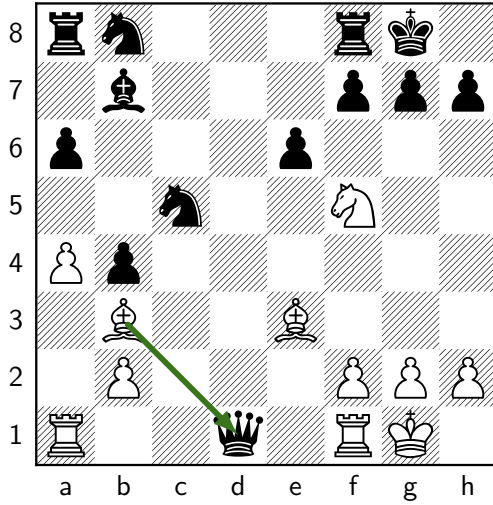
Figure 4: **Move 10, Black: Bxg4.** (a) Our position; Melanie plays Bxg4 (c8g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2639` (ply 35) [structural distance $d=84$, QAP obj=46]: the master played a6; under μ our *Bc1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *pa2* as their *Bd7*, our *nc3* as their *Ph7*; transferred move c8g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2523` (ply 22) [structural distance $d=76$, QAP obj=55]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pa2* as their *Rh1*, our *Nb8* as their *Pa2*; transferred move g8h7. (d) Retrieved from `ccr1_top_p0000#3253` (ply 22) [structural distance $d=76$, QAP obj=55]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pa2* as their *Rh1*, our *Nb8* as their *Pa2*; transferred move g8h7. Synthesis: 1 retrieved analog transferred to Bxg4 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



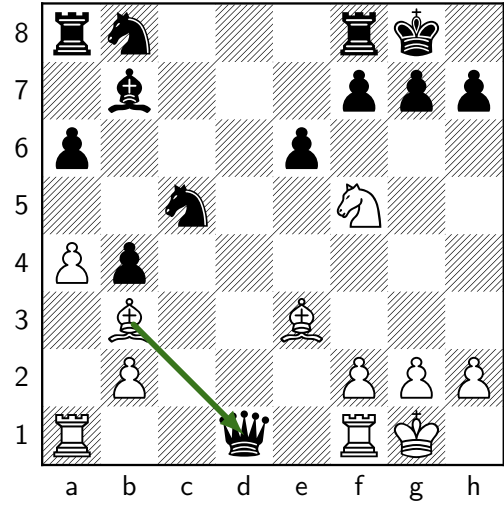
(a)



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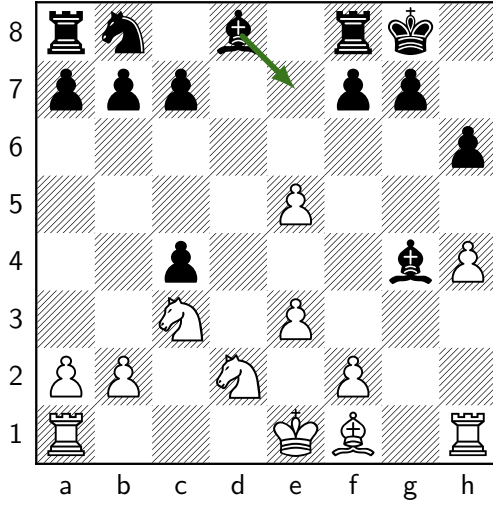


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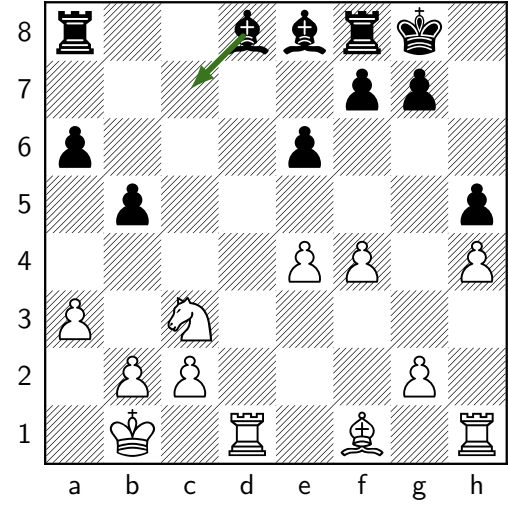


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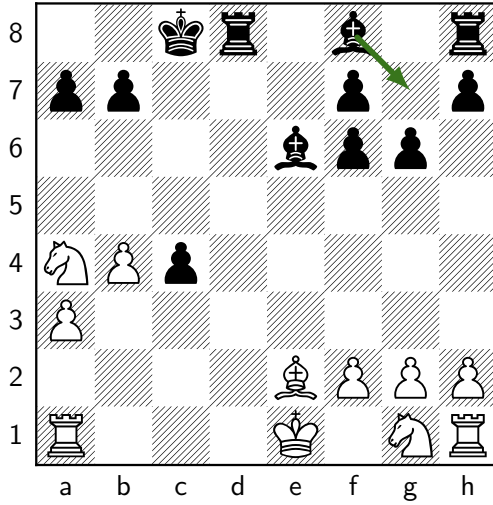
Figure 5: **Move 11, Black: Bxd8**. (a) Our position; Melanie plays Bxd8 (f6d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1718` (ply 34) [structural distance $d=96$, QAP obj=149]: the master played Bxd1; under μ our Bf3 plays the role of that game’s Bb3; cross-type roles: our Ra8 as their Ph2, our Pc7 as their Be3; transferred move f6d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2817` (ply 34) [structural distance $d=96$, QAP obj=149]: the master played Bxd1; under μ our Bf3 plays the role of that game’s Bb3; cross-type roles: our Ra8 as their Ph2, our Pc7 as their Be3; transferred move f6d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1816` (ply 34) [structural distance $d=96$, QAP obj=149]: the master played Bxd1; under μ our Bf3 plays the role of that game’s Bb3; cross-type roles: our Ra8 as their Ph2, our Pc7 as their Be3; transferred move f6d8 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Bxd8 (summed QAP objective 1747, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



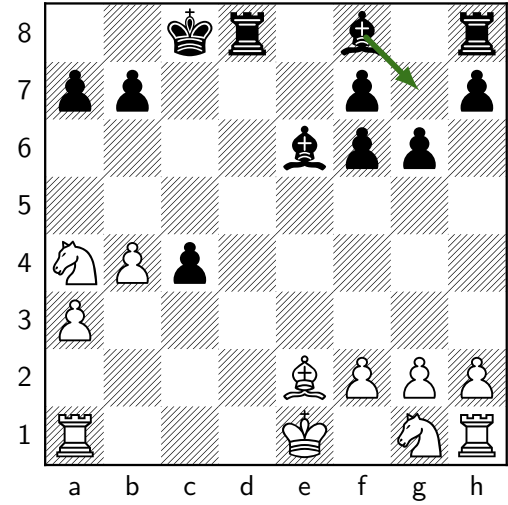
(a)



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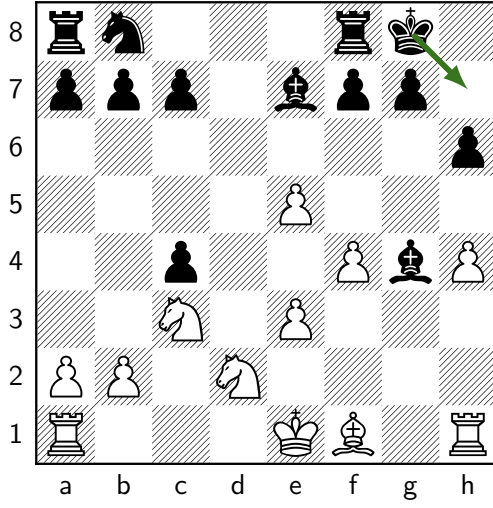


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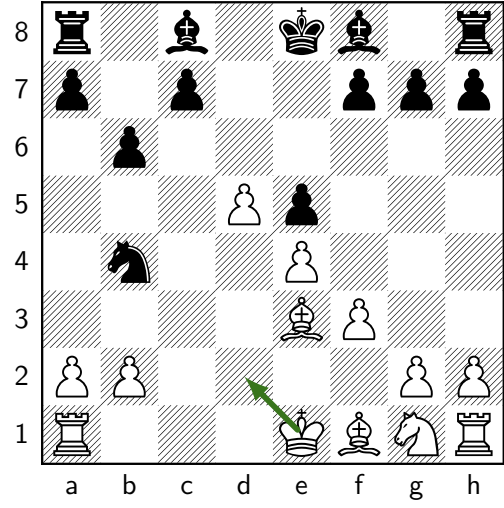


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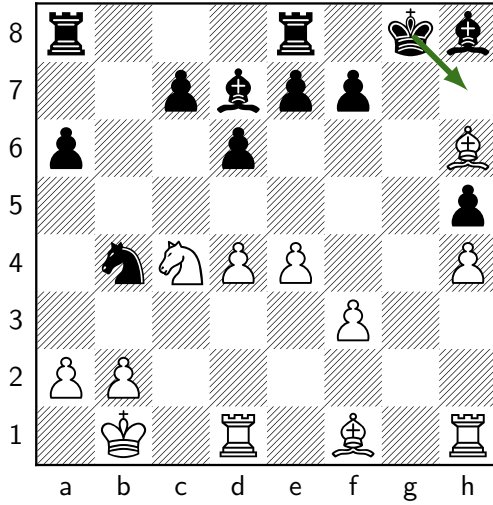
Figure 6: **Move 12, Black: Be7.** (a) Our position; Melanie plays Be7 (d8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#930 (ply 33) [structural distance $d=51$, QAP obj=45]: the master played Bc7; under μ our Bd1 plays the role of that game’s Bd1; cross-type roles: our Bg4 as their Ra8, our pb2 as their Be8; transferred move d8e7 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2503 (ply 27) [structural distance $d=62$, QAP obj=37]: the master played Bg7; under μ our Bd1 plays the role of that game’s Bf1; cross-type roles: our Pa7 as their Rh8, our Bg4 as their Ph7; transferred move d8e7 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2842 (ply 27) [structural distance $d=62$, QAP obj=37]: the master played Bg7; under μ our Bd1 plays the role of that game’s Bf1; cross-type roles: our Pa7 as their Rh8, our Bg4 as their Ph7; transferred move d8e7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Be7 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



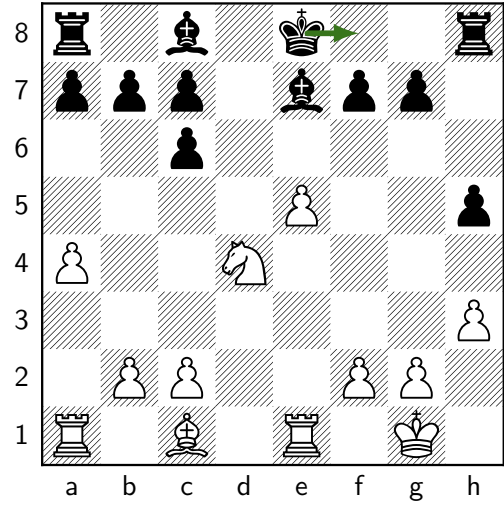
(a)



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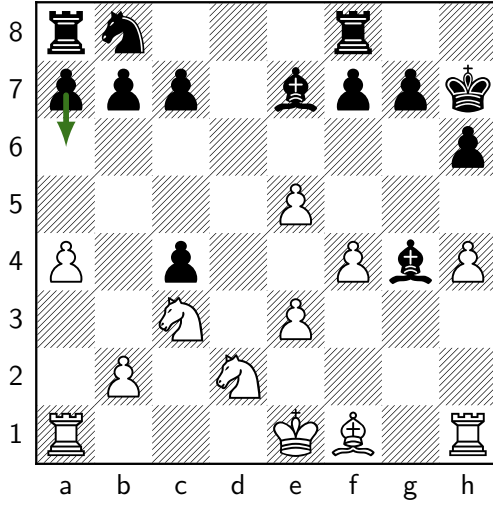


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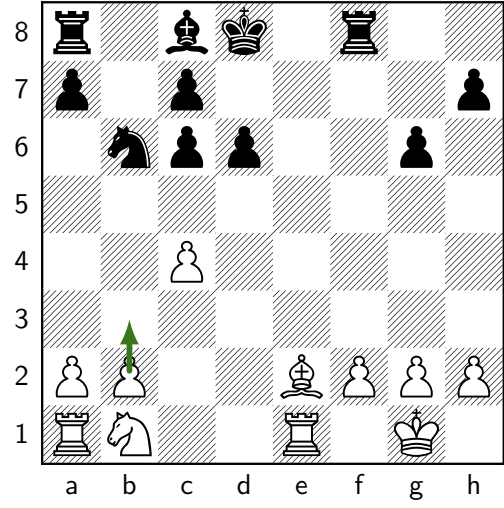


(d)

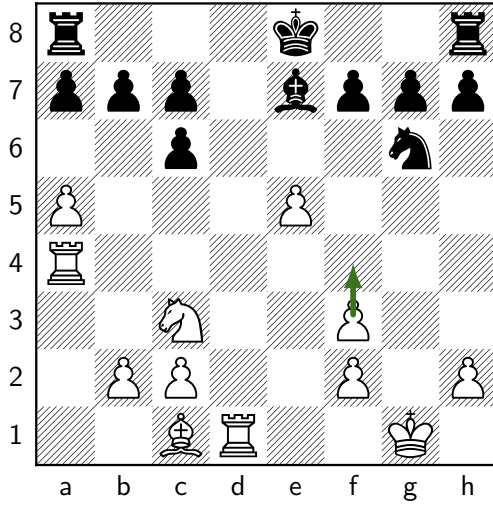
Figure 7: **Move 13, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1906` (ply 22) [structural distance $d=72$, QAP obj=47]: the master played Kd2; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg7* as their *Bf1*, our *Ra8* as their *Pb2*; transferred move g8h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#896` (ply 39) [structural distance $d=63$, QAP obj=40]: the master played Kh7; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Pa7* as their *Ra8*, our *Ph6* as their *Re8*; transferred move g8h7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#125` (ply 29) [structural distance $d=75$, QAP obj=30]: the master played Kf8; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *nc3* as their *Ra8*, our *Ph6* as their *Be7*; transferred move g8h7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh7 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



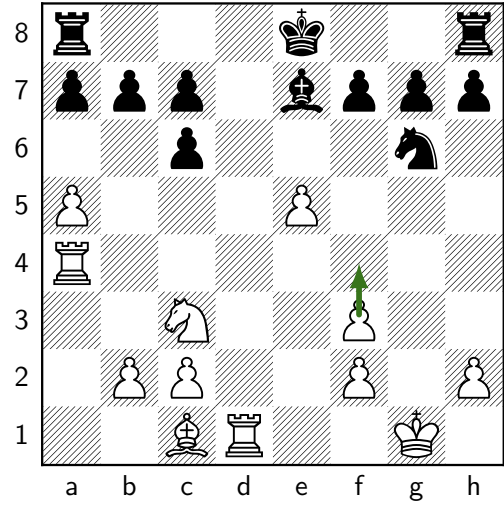
(a)



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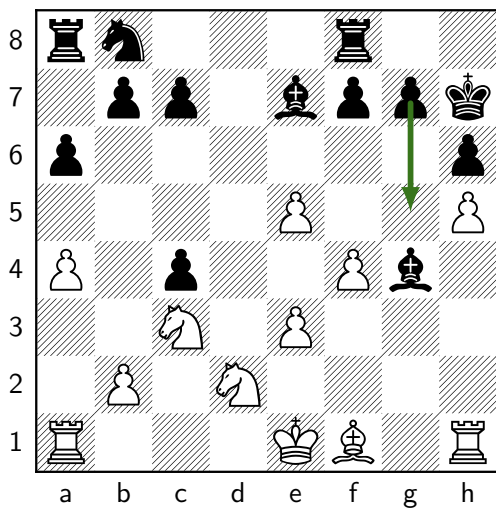


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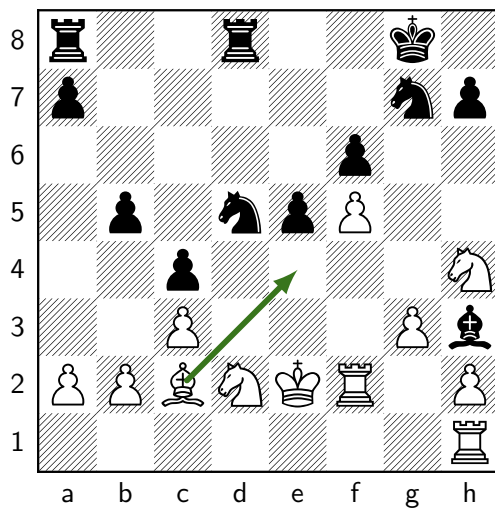


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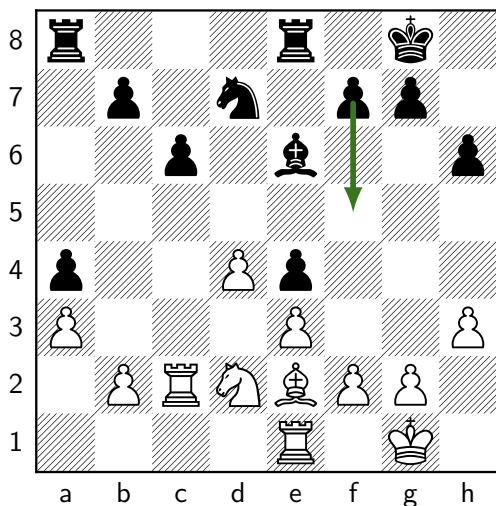
Figure 8: **Move 14, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2807` (ply 38) [structural distance $d=65$, QAP obj=29]: the master played b3; under μ our $Pa2$ plays the role of that game’s $Pb2$; cross-type roles: our $Nb8$ as their $Ra1$, our $Pf7$ as their $Nb1$; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2256` (ply 28) [structural distance $d=70$, QAP obj=25]: the master played f4; under μ our $Pa2$ plays the role of that game’s $Pf3$; cross-type roles: our $pf4$ as their $Bc1$, our $Pf7$ as their $Rd1$; transferred move a7a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2739` (ply 28) [structural distance $d=70$, QAP obj=25]: the master played f4; under μ our $Pa2$ plays the role of that game’s $Pf3$; cross-type roles: our $pf4$ as their $Bc1$, our $Pf7$ as their $Rd1$; transferred move a7a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a6 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



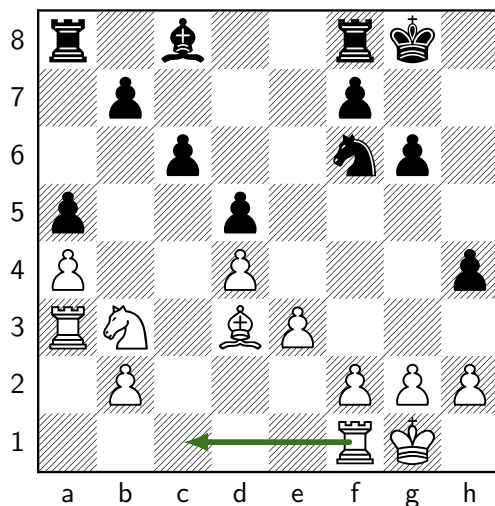
(a)



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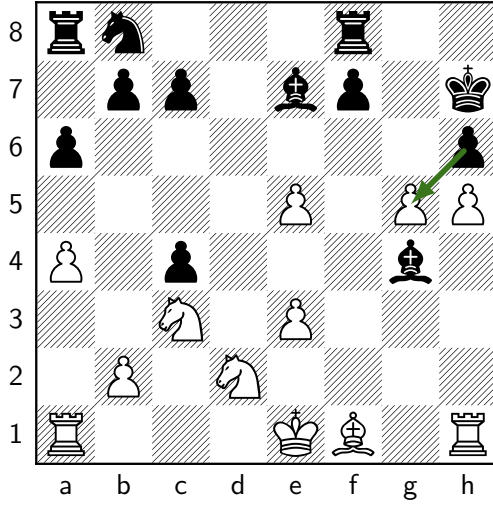


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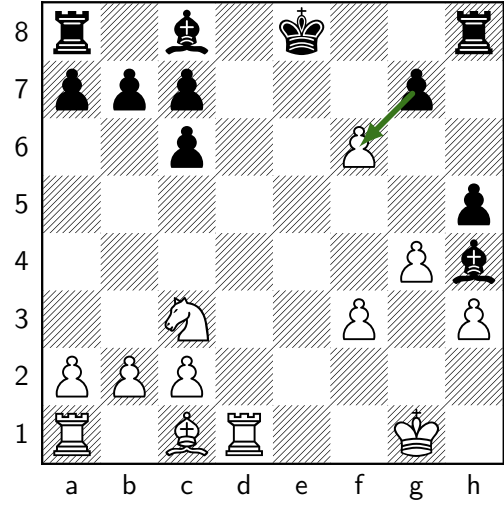


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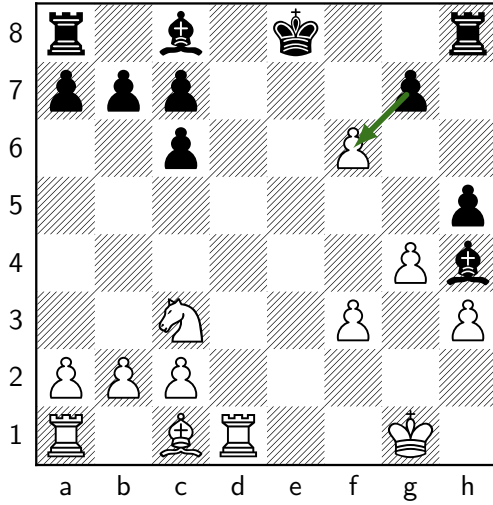
Figure 9: **Move 15, Black: g5**. (a) Our position; Melanie plays g5 (g7g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3362` (ply 44) [structural distance $d=65$, QAP obj=34]: the master played Be4; under μ our Pg2 plays the role of that game’s Bc2 (a **cross-type** correspondence); cross-type roles: our Nb8 as their Rh1, our Bg4 as their Nd2; transferred move g7g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3307` (ply 43) [structural distance $d=62$, QAP obj=20]: the master played f5; under μ our Pg2 plays the role of that game’s Pf2; cross-type roles: our Nb8 as their Ra8, our Be7 as their Nd7; transferred move g7g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3939` (ply 34) [structural distance $d=69$, QAP obj=20]: the master played Rc1; under μ our Pg2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ra8 as their Pf2, our bf1 as their Pd4; transferred move g7g5 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to g5 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



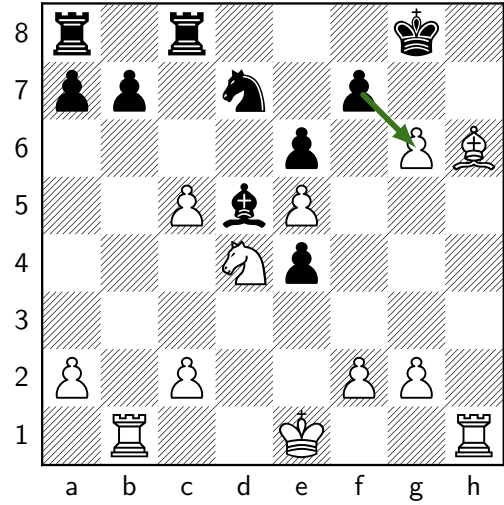
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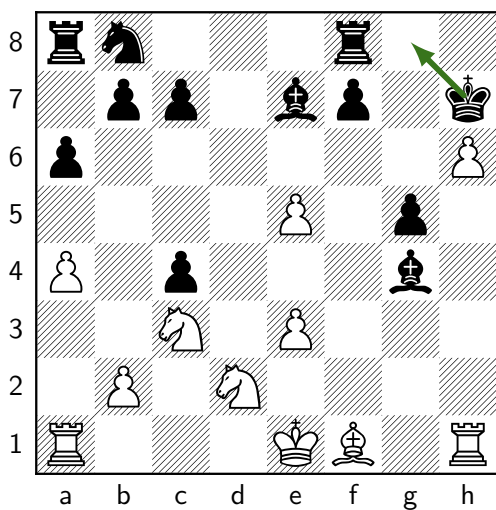


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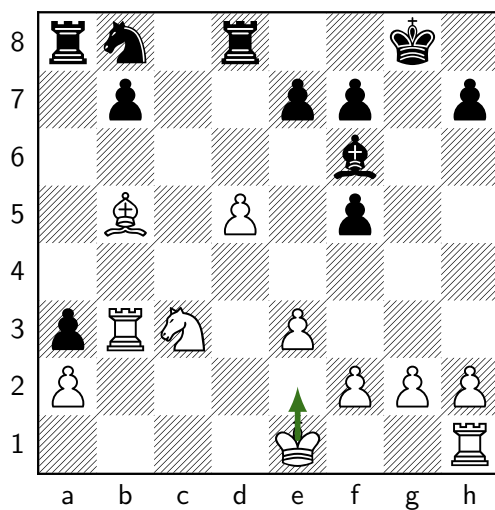


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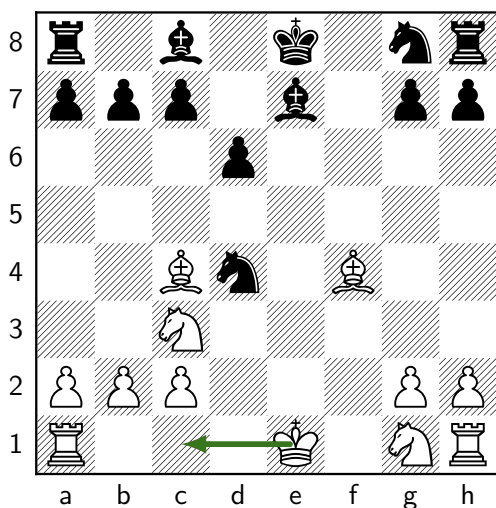
Figure 10: **Move 16, Black: hxg5.** (a) Our position; Melanie plays hxg5 (h6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1436 (ply 29) [structural distance $d=71$, QAP obj=86]: the master played gxf6; under μ our Ph3 plays the role of that game’s Pg2; cross-type roles: our Pf7 as their Rh8, our Nb8 as their Pa7; transferred move h6g5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3141 (ply 29) [structural distance $d=71$, QAP obj=86]: the master played gxf6; under μ our Ph3 plays the role of that game’s Pg2; cross-type roles: our Pf7 as their Rh8, our Nb8 as their Pa7; transferred move h6g5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3592 (ply 43) [structural distance $d=80$, QAP obj=78]: the master played fxg6; under μ our Ph3 plays the role of that game’s Pf2; cross-type roles: our bf1 as their Rc8, our nc3 as their Pa7; transferred move h6g5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to hxg5 (summed QAP objective 688, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



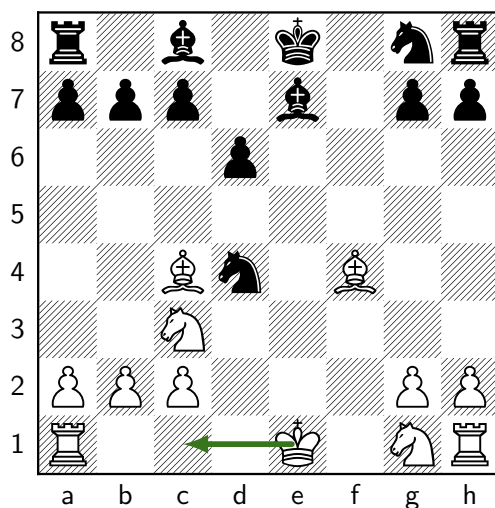
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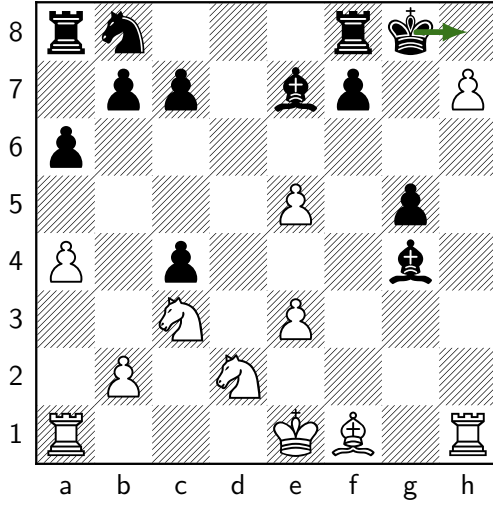


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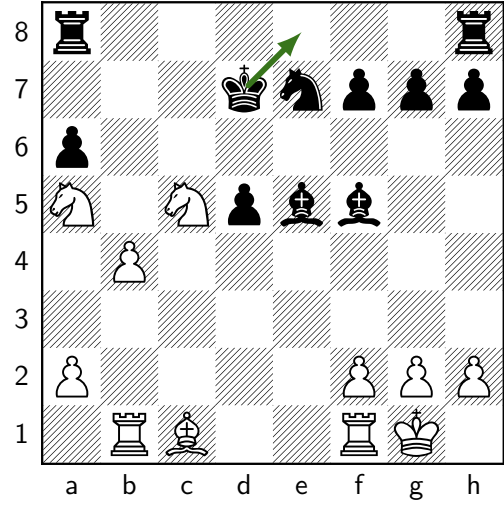


(d)

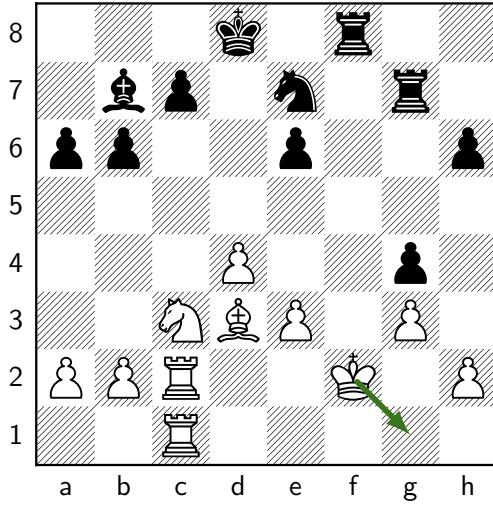
Figure 11: **Move 17, Black: Kg8 (h7g8)**. (a) Our position; Melanie plays Kg8 (h7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#586` (ply 36) [structural distance $d=70$, QAP obj=26]: the master played Ke2; under μ our Kh2 plays the role of that game’s Ke1; cross-type roles: our Nb8 as their Pg2, our Rf8 as their Ph2; transferred move h7g8 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#432` (ply 18) [structural distance $d=67$, QAP obj=19]: the master played 0-0-0; under μ our Kh2 plays the role of that game’s Ke1; cross-type roles: our Nb8 as their Rh1, our Rf8 as their Pa2; transferred move h7g8 (**endorses this move**). (d) Retrieved from `ccl1_top_p0000#433` (ply 18) [structural distance $d=67$, QAP obj=19]: the master played 0-0-0; under μ our Kh2 plays the role of that game’s Ke1; cross-type roles: our Nb8 as their Rh1, our Rf8 as their Pa2; transferred move h7g8 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg8 (summed QAP objective 228, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



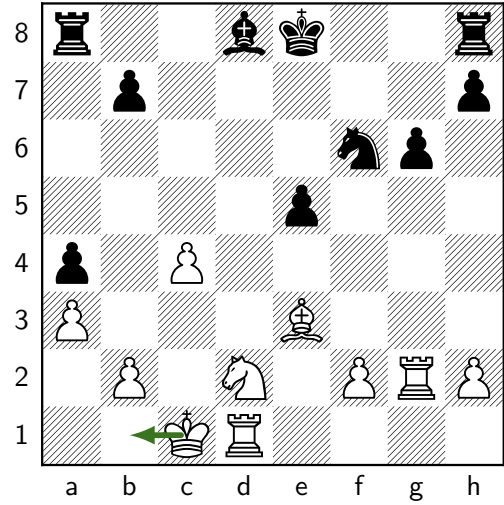
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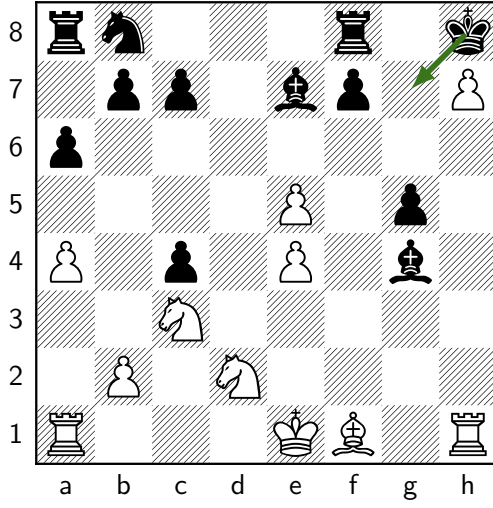


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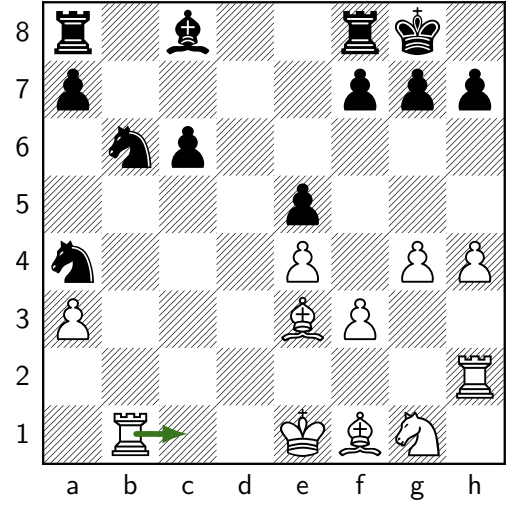


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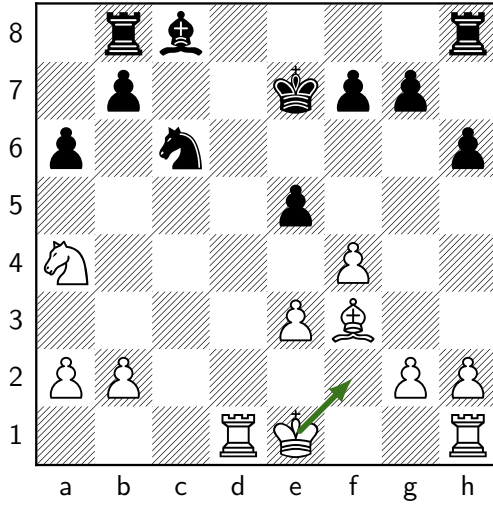
Figure 12: **Move 18, Black: Kh8.** (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#437` (ply 39) [structural distance $d=72$, QAP obj=38]: the master played Ke8; under μ our Kg1 plays the role of that game's Kd2; cross-type roles: our Nb8 as their Ra8, our Pb7 as their Rh8; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1820` (ply 42) [structural distance $d=72$, QAP obj=16]: the master played Kg1; under μ our Kg1 plays the role of that game's Kf2; cross-type roles: our bf1 as their Rc1, our Nb8 as their Pa2; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3232` (ply 46) [structural distance $d=71$, QAP obj=4]: the master played Kb1; under μ our Kg1 plays the role of that game's Kc1; cross-type roles: our Bg4 as their Nd2, our bf1 as their Pf2; transferred move g8h8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh8 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



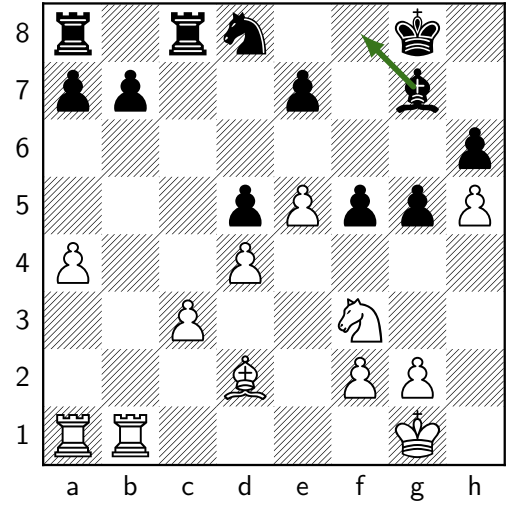
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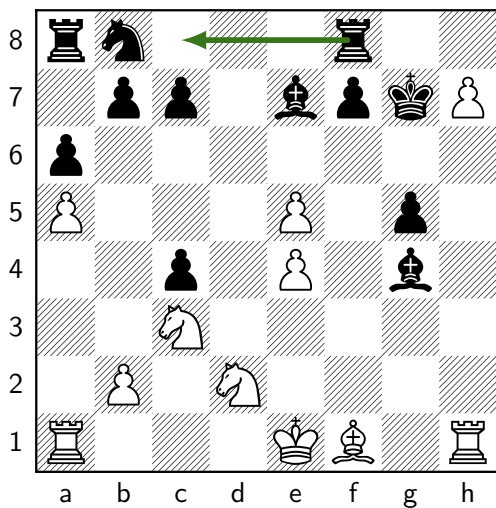


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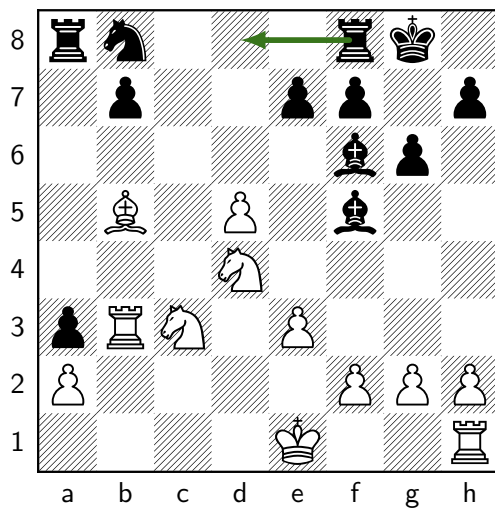


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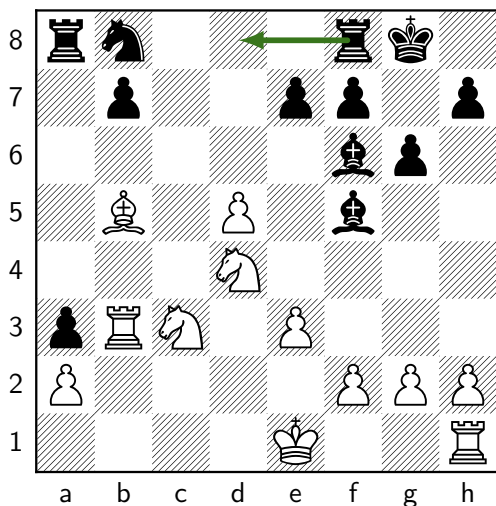
Figure 13: **Move 19, Black: Kg7.** (a) Our position; Melanie plays Kg7 (h8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3548` (ply 34) [structural distance $d=77$, QAP obj=28]: the master played Rc1; under μ our Kh1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Bf1, our Pf7 as their Ng1; transferred move h8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2388` (ply 32) [structural distance $d=72$, QAP obj=26]: the master played Kf2; under μ our Kh1 plays the role of that game’s Ke1; cross-type roles: our bf1 as their Rd1, our Ra8 as their Pa2; transferred move h8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1975` (ply 47) [structural distance $d=78$, QAP obj=25]: the master played Bf8; under μ our Kh1 plays the role of that game’s Bg2 (a **cross-type** correspondence); cross-type roles: our pe4 as their Ra8, our Bg4 as their Rc8; transferred move h8g7 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg7 (summed QAP objective 254, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



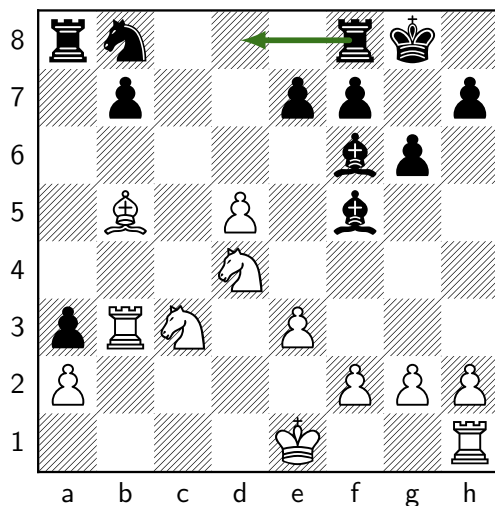
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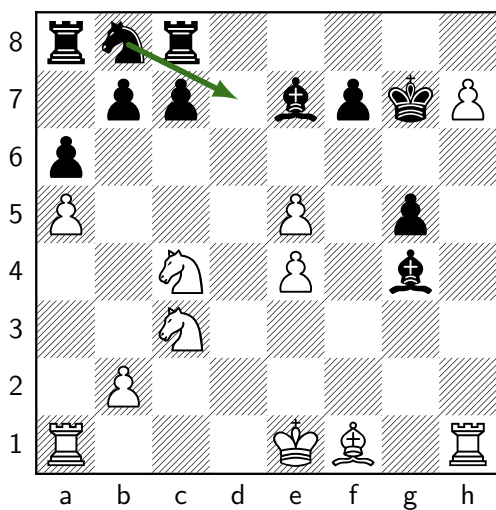


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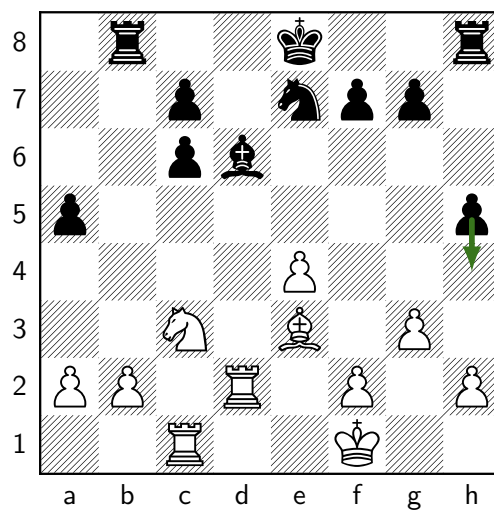


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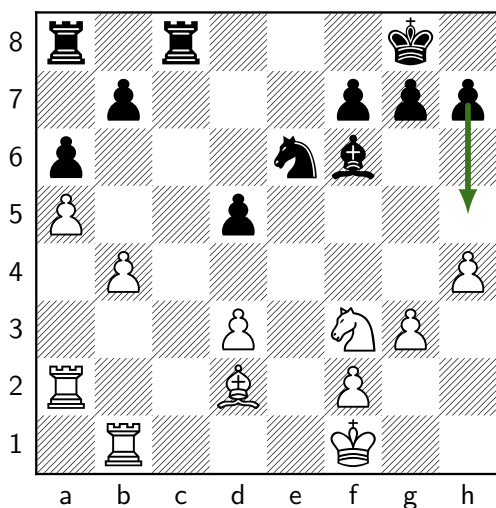
Figure 14: **Move 20, Black: Rc8**. (a) Our position; Melanie plays Rc8 (f8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#115` (ply 33) [structural distance $d=80$, QAP obj=16]: the master played Rd8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pc7 as their Nb8, our bf1 as their Pe7; transferred move f8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1466` (ply 33) [structural distance $d=80$, QAP obj=16]: the master played Rd8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pc7 as their Nb8, our bf1 as their Pe7; transferred move f8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#586` (ply 33) [structural distance $d=80$, QAP obj=16]: the master played Rd8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our Pc7 as their Nb8, our bf1 as their Pe7; transferred move f8c8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rc8 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



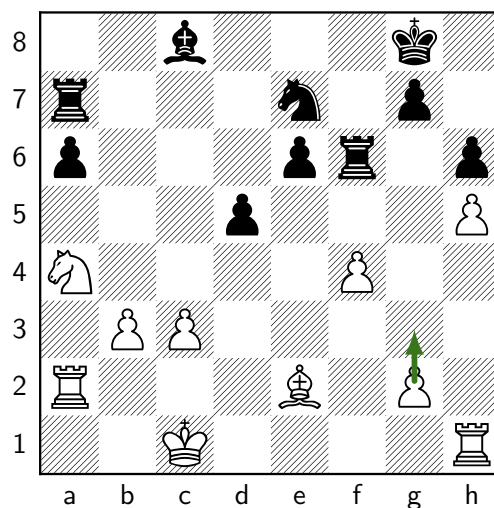
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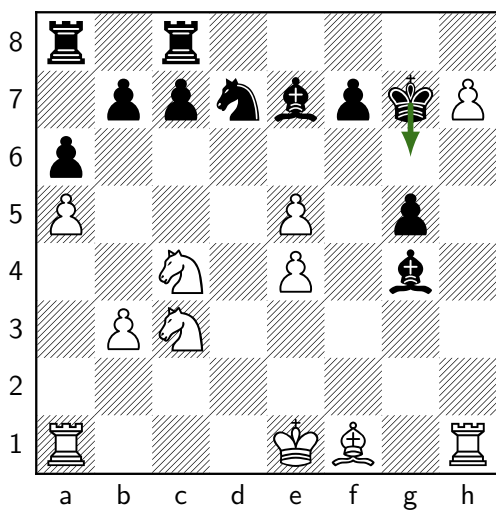


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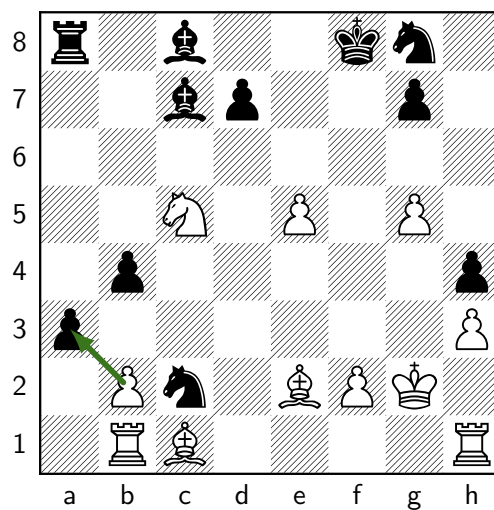


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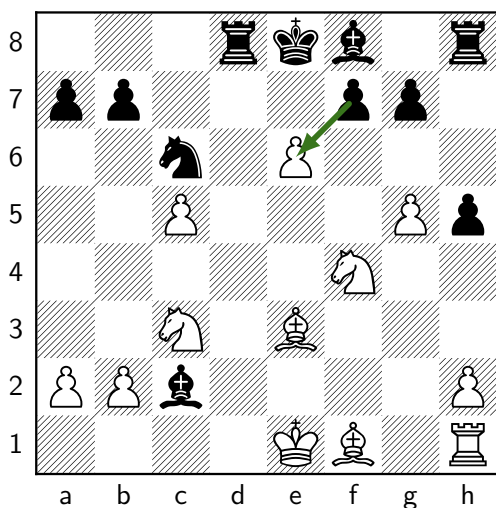
Figure 15: **Move 21, Black: Nd7.** (a) Our position; Melanie plays Nd7 (b8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1729` (ply 33) [structural distance $d=84$, QAP obj=13]: the master played h4; under μ our Nb1 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our Bg4 as their Rb8, our pe5 as their Rh8; transferred move b8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#744` (ply 51) [structural distance $d=70$, QAP obj=11]: the master played h5; under μ our Nb1 plays the role of that game’s Ph2 (a **cross-type** correspondence); cross-type roles: our nc4 as their Ra8, our nc3 as their Rc8; transferred move b8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1302` (ply 50) [structural distance $d=81$, QAP obj=10]: the master played g3; under μ our Nb1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our pe5 as their Ra2, our rh1 as their Be2; transferred move b8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd7 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



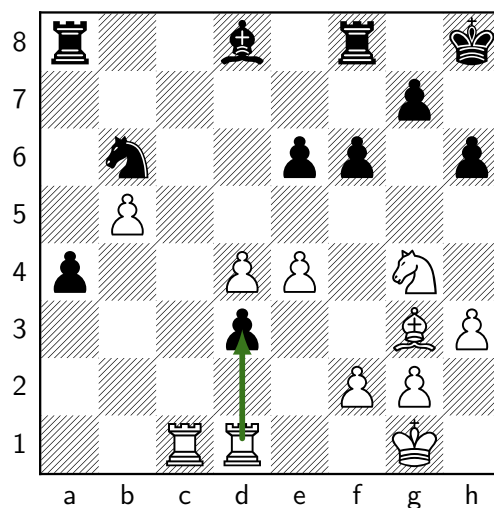
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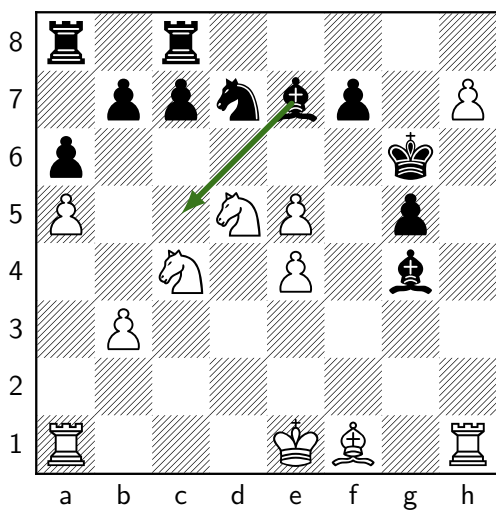


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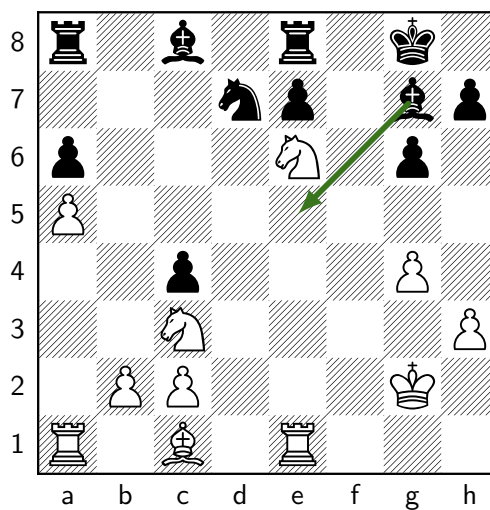


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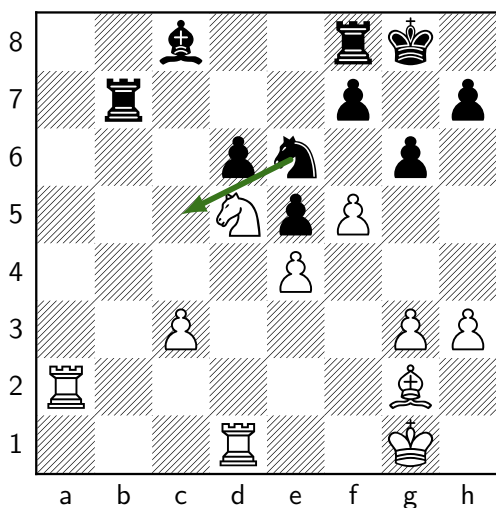
Figure 16: **Move 22, Black: Kg6**. (a) Our position; Melanie plays Kg6 (**g7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3854` (ply 50) [structural distance $d=79$, QAP obj=26]: the master played **bxa3**; under μ our *Kg2* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *pe4* as their *Rb1*, our *Rc8* as their *Bc1*; transferred move **g7g6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2331` (ply 33) [structural distance $d=80$, QAP obj=26]: the master played **fxe6**; under μ our *Kg2* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *nc4* as their *Rd8*, our *Pc7* as their *Bf8*; transferred move **g7g6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1458` (ply 56) [structural distance $d=69$, QAP obj=24]: the master played **Rxd3**; under μ our *Kg2* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Rc1*, our *Bg4* as their *Ph3*; transferred move **g7g6** (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kg6 (summed QAP objective 182, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



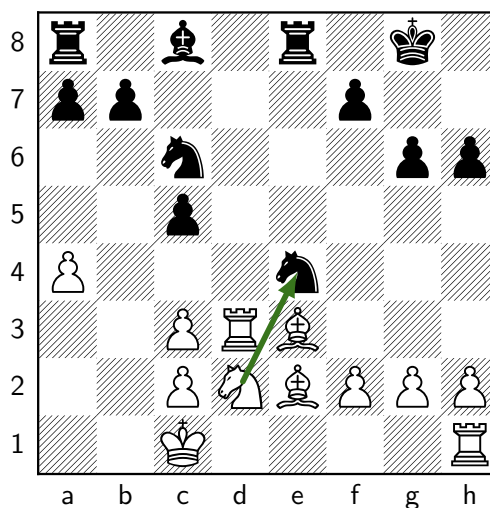
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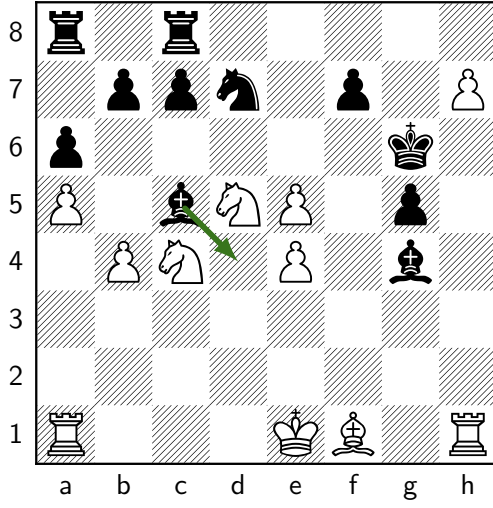


(c)

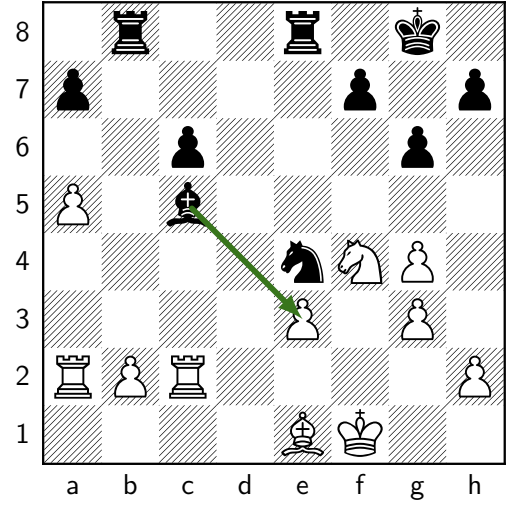


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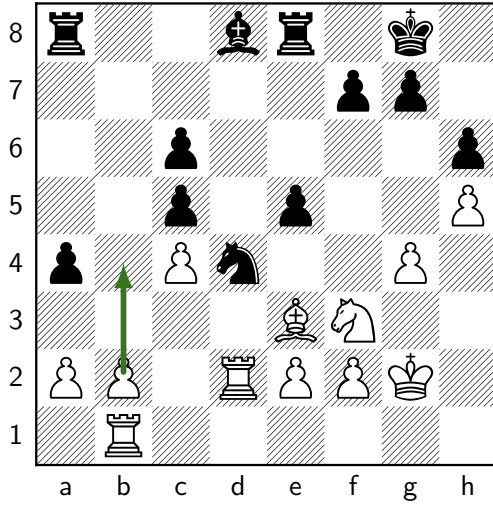
Figure 17: **Move 23, Black: Bc5.** (a) Our position; Melanie plays Bc5 (e7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#3819` (ply 49) [structural distance $d=74$, QAP obj=22]: the master played Be5; under μ our Be2 plays the role of that game’s Bg2; cross-type roles: our pa5 as their Ra8, our Pa6 as their Bc8; transferred move e7c5 (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#12` (ply 53) [structural distance $d=84$, QAP obj=18]: the master played Nc5; under μ our Be2 plays the role of that game’s Ne3 (a **cross-type** correspondence); cross-type roles: our Rc8 as their Bc8, our Pf7 as their Rf8; transferred move e7c5 (**endorses this move**). (d) Retrieved from `ccr1.top_p0003#1429` (ply 34) [structural distance $d=81$, QAP obj=44]: the master played Nxe4; under μ our Be2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Pb7 as their Rh1, our Ra8 as their Be2; transferred move e7d6. Synthesis: 2 retrieved analogs transferred to Bc5 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



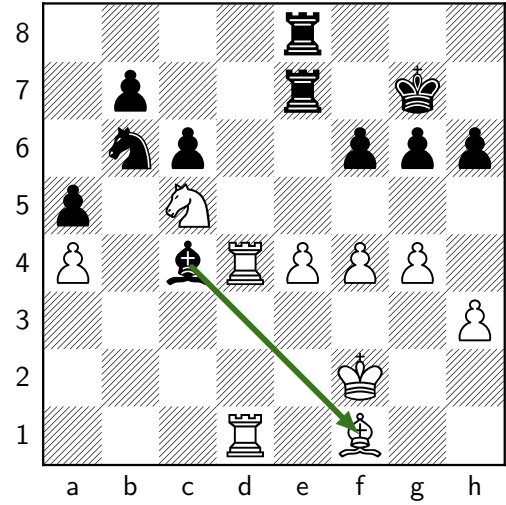
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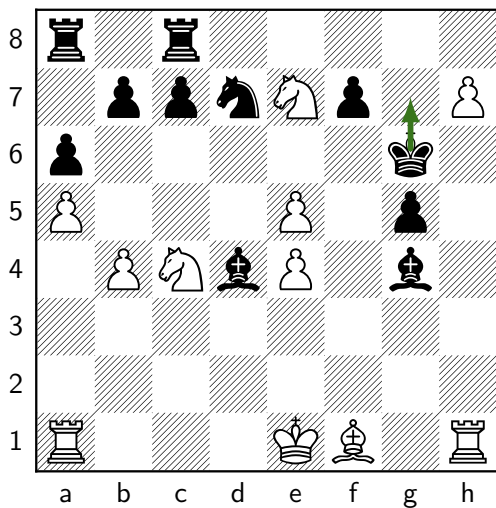


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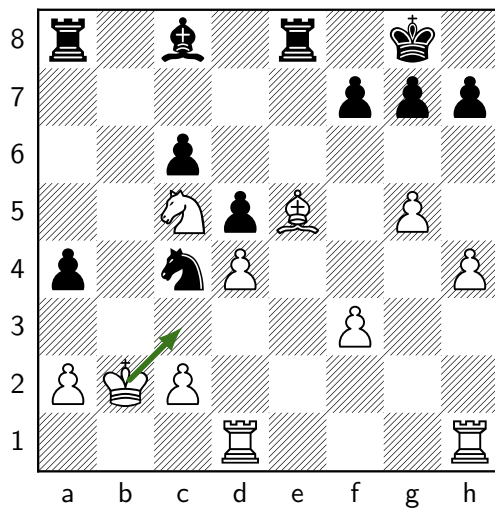


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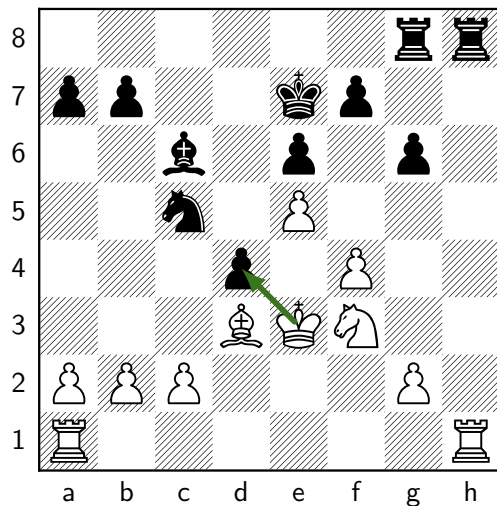
Figure 18: **Move 24, Black: Bd4.** (a) Our position; Melanie plays Bd4 (c5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2494` (ply 55) [structural distance $d=79$, QAP obj=10]: the master played Bxe3; under μ our Bc4 plays the role of that game’s Bc4; cross-type roles: our nd5 as their Rb8, our Pf7 as their Re8; transferred move c5d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3799` (ply 52) [structural distance $d=87$, QAP obj=37]: the master played b4; under μ our Pb2 plays the role of that game’s Pb2; cross-type roles: our Rc8 as their Pa2, our pb4 as their Rd2; transferred move b7b5. (d) Retrieved from `ccr1_top_p0003#2289` (ply 59) [structural distance $d=77$, QAP obj=30]: the master played Bxf1; under μ our Bc4 plays the role of that game’s Bc5; cross-type roles: our Bg4 as their Re8, our Pf7 as their Re7; transferred move c5f2. Synthesis: 1 retrieved analog transferred to Bd4 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



(a)

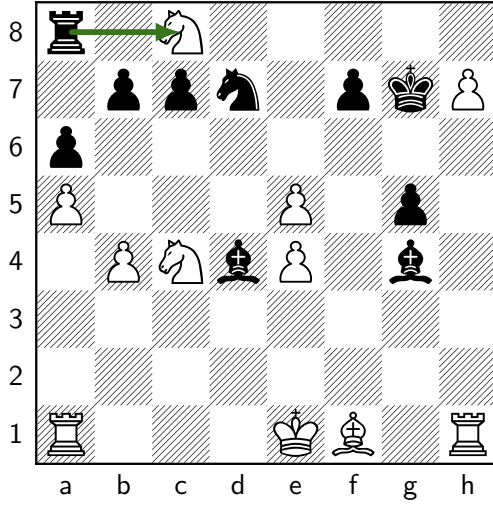


(b)

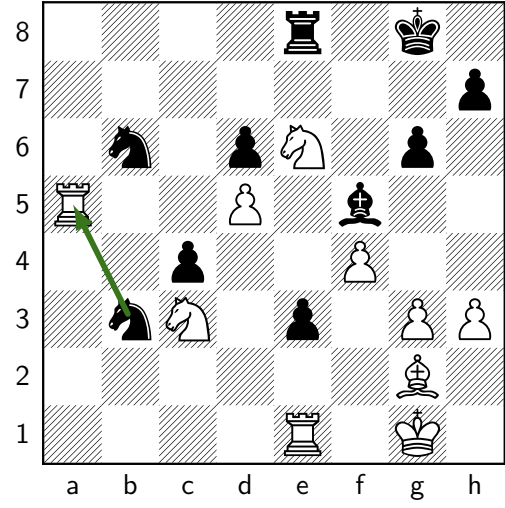


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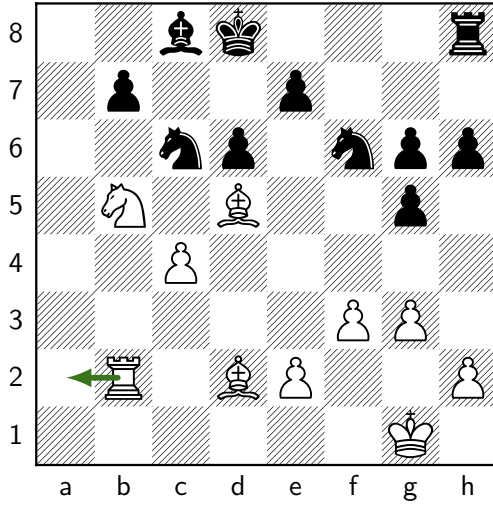
Figure 19: **Move 25, Black: Kg7.** (a) Our position; Melanie plays Kg7 (**g6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1845` (ply 54) [structural distance $d=70$, QAP obj=84]: the master played Kc3; under μ our Kg3 plays the role of that game’s Kb2; cross-type roles: our pb4 as their Rd1, our Nd7 as their Pa2; transferred move g6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2258` (ply 40) [structural distance $d=83$, QAP obj=37]: the master played Kxd4; under μ our Kg3 plays the role of that game’s Ke3; cross-type roles: our pe5 as their Rh1, our Nd7 as their Pc2; transferred move g6g7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg7 (summed QAP objective 135, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



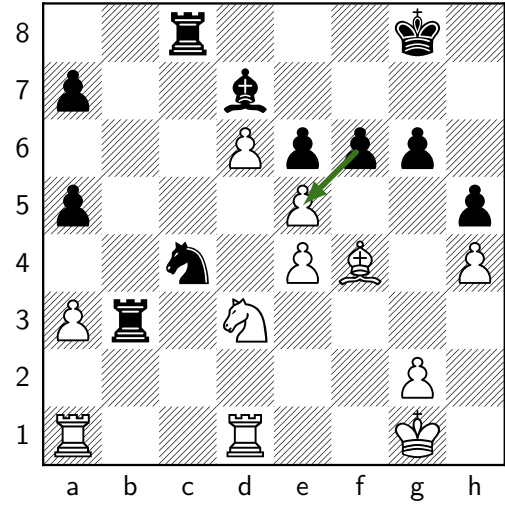
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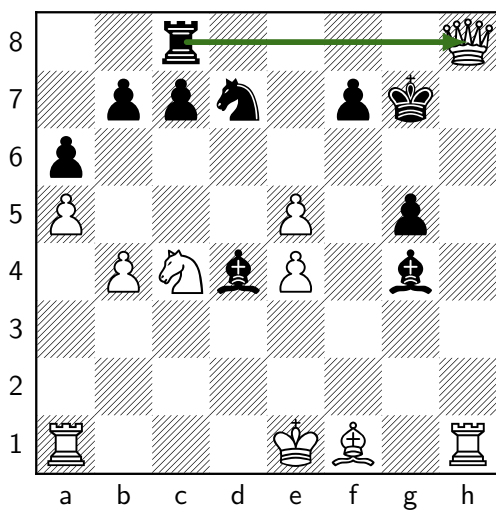


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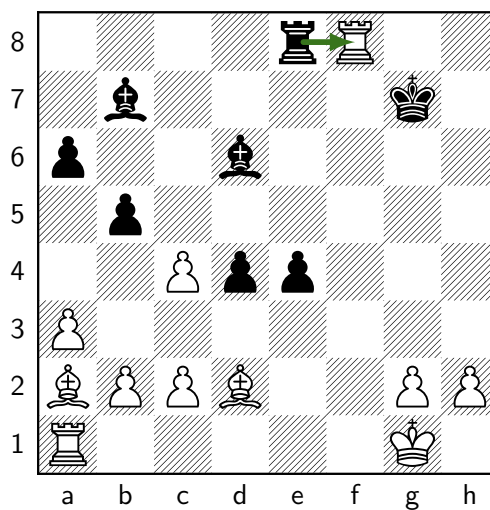


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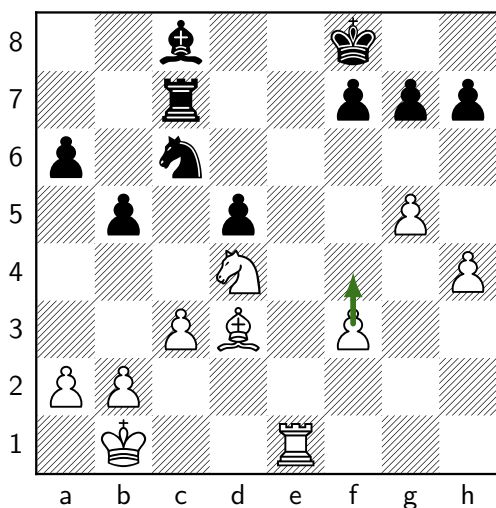
Figure 20: **Move 26, Black: Rxc8.** (a) Our position; Melanie plays Rxc8 (a8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#67 (ply 65) [structural distance $d=86$, QAP obj=27]: the master played Nxa5; under μ our *Ra1* plays the role of that game’s *Nb6* (a **cross-type** correspondence); cross-type roles: our *Bd4* as their *Re8*, our *rh1* as their *Nb6*; transferred move a8c8 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#498 (ply 48) [structural distance $d=85$, QAP obj=16]: the master played Ra2; under μ our *Ra1* plays the role of that game’s *Rb2*; cross-type roles: our *rh1* as their *Bd2*, our *nc8* as their *Pe2*; transferred move a8c8 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#172 (ply 55) [structural distance $d=87$, QAP obj=48]: the master played fxe5; under μ our *Bd5* plays the role of that game’s *Pf3* (a **cross-type** correspondence); cross-type roles: our *Bg4* as their *Rc8*, our *nc8* as their *Pa7*; transferred move d4e5. Synthesis: 2 retrieved analogs transferred to Rxc8 (summed QAP objective 43, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



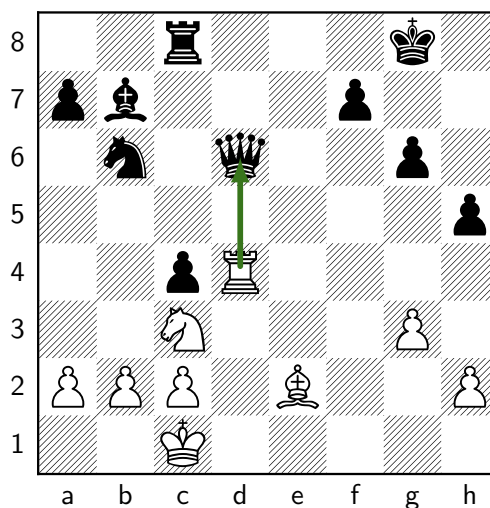
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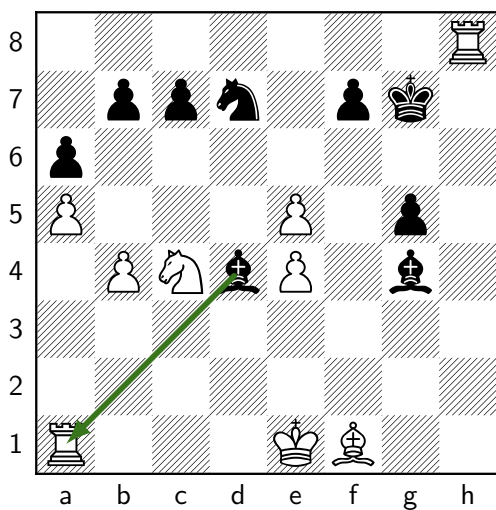


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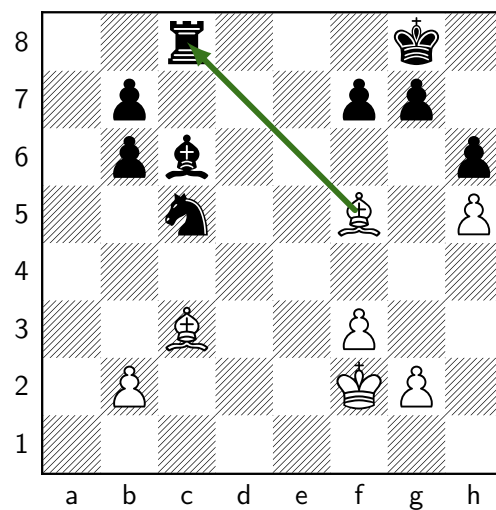


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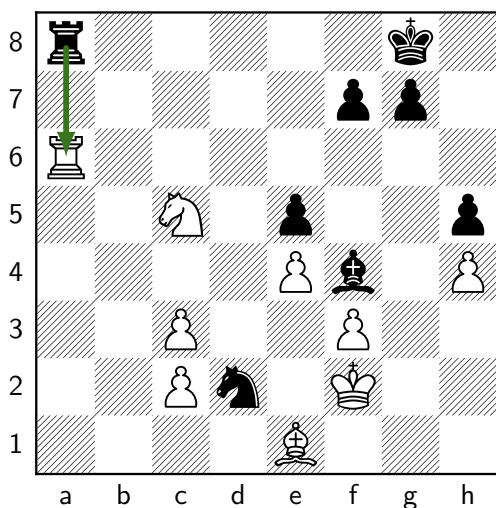
Figure 21: **Move 27, Black: Rxf8.** (a) Our position; Melanie plays Rxf8 (c8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#87` (ply 49) [structural distance $d=98$, QAP obj=7]: the master played Rxf8; under μ our *Rc1* plays the role of that game’s *Re1*; cross-type roles: our *qh8* as their *rf8*, our *Nd7* as their *Pe4*; transferred move c8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2788` (ply 58) [structural distance $d=99$, QAP obj=6]: the master played f4; under μ our *Rc1* plays the role of that game’s *Pf3* (a **cross-type** correspondence); cross-type roles: our *Bg4* as their *Re1*, our *Bd4* as their *Pc3*; transferred move c8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1593` (ply 62) [structural distance $d=87$, QAP obj=6]: the master played Rxd6; under μ our *Rc1* plays the role of that game’s *Rd4*; cross-type roles: our *bf1* as their *Pa2*, our *nc4* as their *Be2*; transferred move c8h8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxf8 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



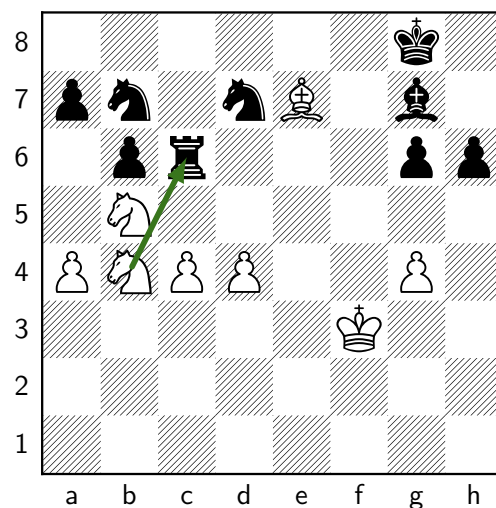
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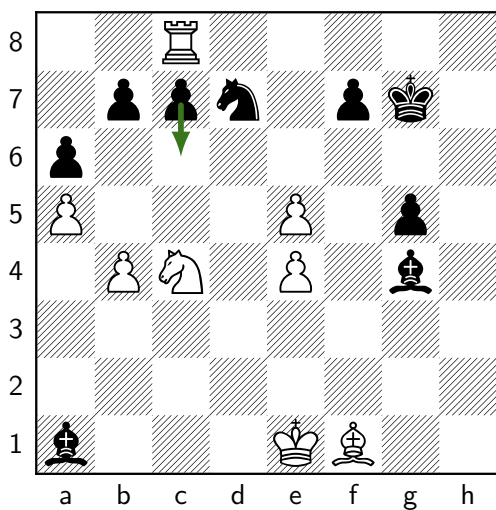


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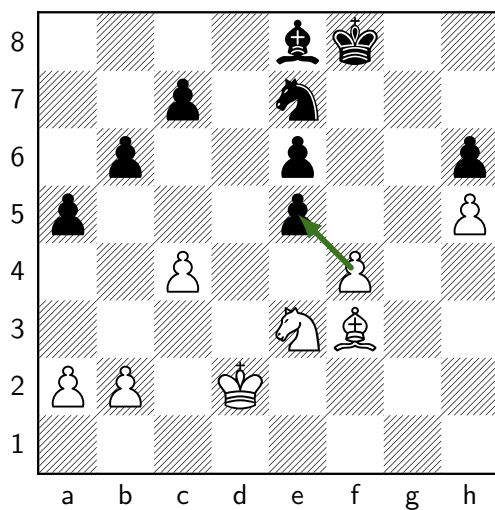


(d)

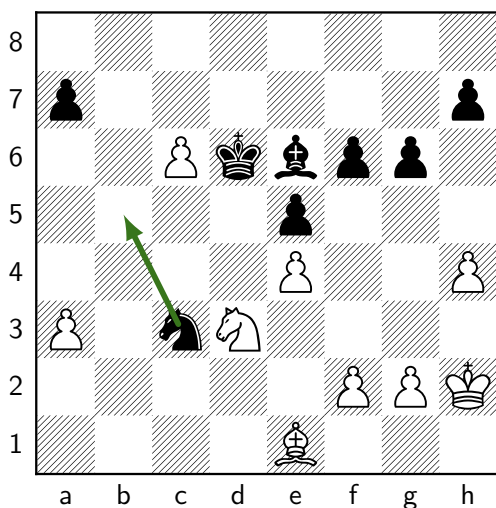
Figure 22: **Move 28, Black: Bxa1**. (a) Our position; Melanie plays **Bxa1** (**d4a1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3838` (ply 70) [structural distance $d=64$, QAP obj=15]: the master played **Bxc8**; under μ our **Bd5** plays the role of that game’s **Bf5**; cross-type roles: our **Pa6** as their **Bc3**, our **Nd7** as their **Ph5**; transferred move **d4a1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1690` (ply 59) [structural distance $d=68$, QAP obj=15]: the master played **Rxa6**; under μ our **Bd5** plays the role of that game’s **Ra1** (a **cross-type** correspondence); cross-type roles: our **Nd7** as their **pe4**, our **rh8** as their **Nd2**; transferred move **d4a1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#690` (ply 66) [structural distance $d=50$, QAP obj=11]: the master played **Nxc6**; under μ our **Bd5** plays the role of that game’s **Nb4** (a **cross-type** correspondence); cross-type roles: our **Nd7** as their **Pd4**, our **pb4** as their **Nb5**; transferred move **d4a1** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **Bxa1** (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



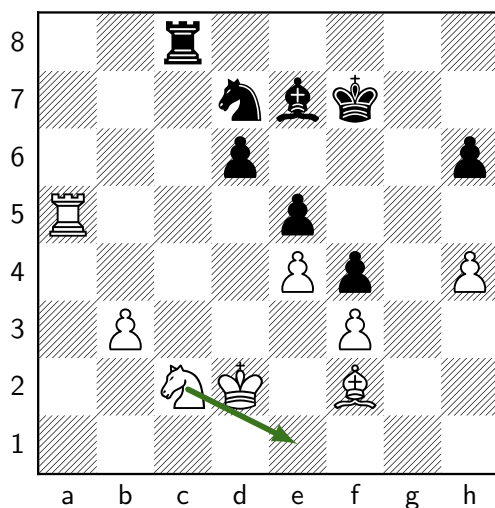
(a)



(b)

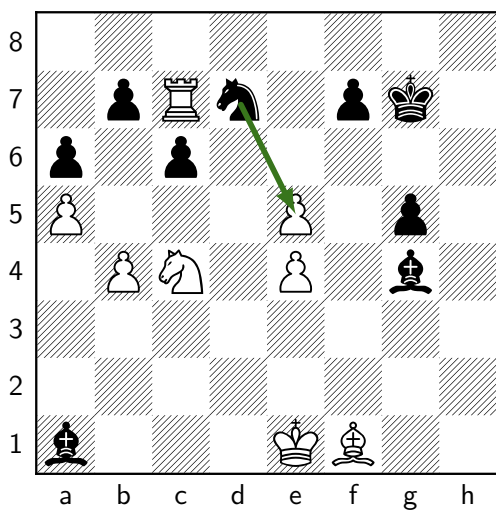


(c)

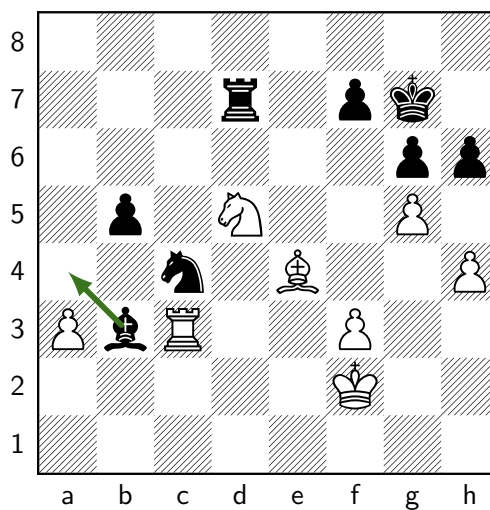


(d)

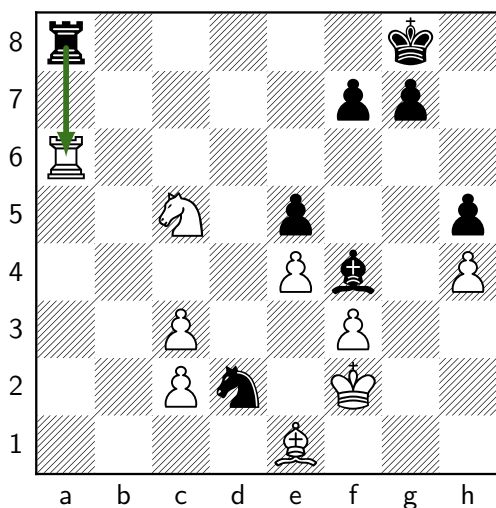
Figure 23: **Move 29, Black: c6**. (a) Our position; Melanie plays c6 (c7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#390 (ply 72) [structural distance $d=45$, QAP obj=29]: the master played fxe5; under μ our Pc2 plays the role of that game’s Pf4; cross-type roles: our Bg4 as their Ne3, our rc8 as their pe5; transferred move c7c6 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2293 (ply 69) [structural distance $d=39$, QAP obj=22]: the master played Nb5; under μ our Pc2 plays the role of that game’s Nc6 (a **cross-type** correspondence); cross-type roles: our Ba1 as their Ph7, our nc4 as their pa3; transferred move c7c6 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1642 (ply 66) [structural distance $d=34$, QAP obj=19]: the master played Ne1; under μ our Pc2 plays the role of that game’s Nc2 (a **cross-type** correspondence); cross-type roles: our pa5 as their Bf2, our Nd7 as their Pe4; transferred move c7c6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to c6 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



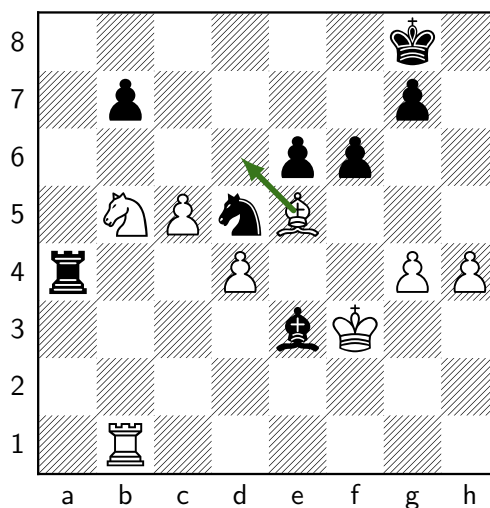
(a)



(b)

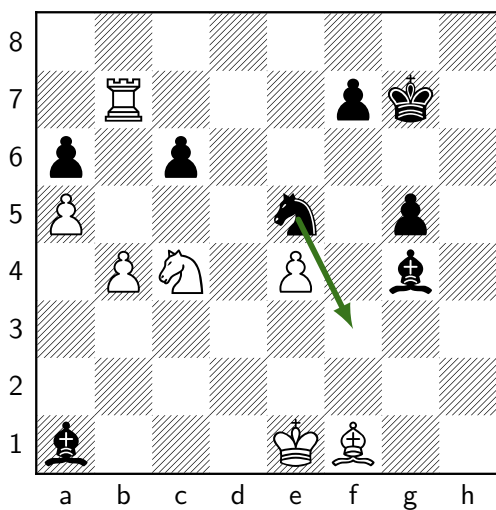


(c)

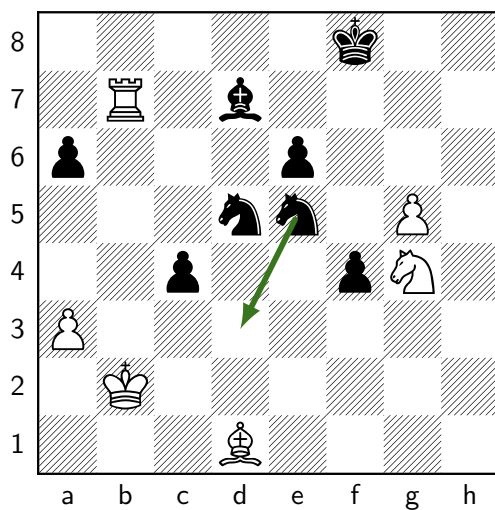


(d)

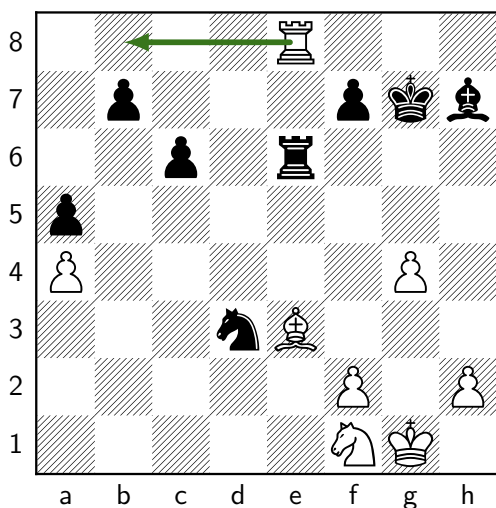
Figure 24: **Move 30, Black: Nxe5**. (a) Our position; Melanie plays Nxe5 (d7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2147` (ply 73) [structural distance $d=51$, QAP obj=49]: the master played Ba4; under μ our Nd2 plays the role of that game’s Bb6 (a **cross-type** correspondence); cross-type roles: our Pb7 as their nd5, our Pc6 as their Nc4; transferred move d7e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1690` (ply 59) [structural distance $d=53$, QAP obj=29]: the master played Rxa6; under μ our Nd2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our bf1 as their pe4, our rc7 as their be1; transferred move d7e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1776` (ply 72) [structural distance $d=46$, QAP obj=28]: the master played Bd6; under μ our Nd2 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our pe4 as their Rb1, our rc7 as their be3; transferred move d7e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nxe5 (summed QAP objective 157, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



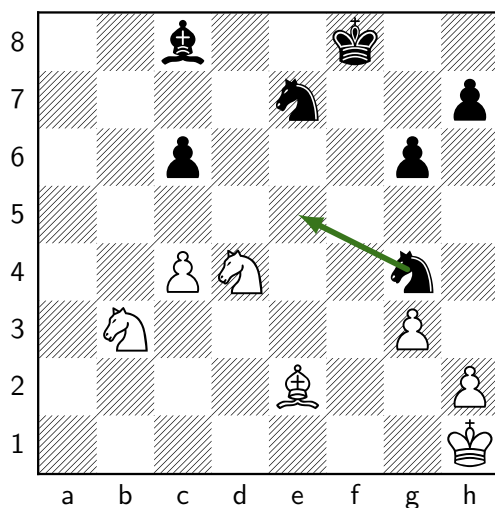
(a)



(b)

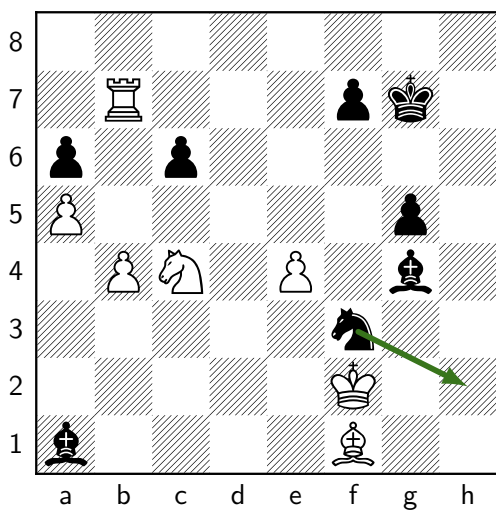


(c)

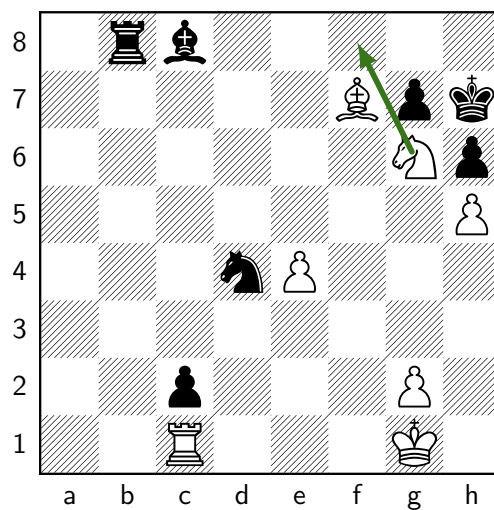


(d)

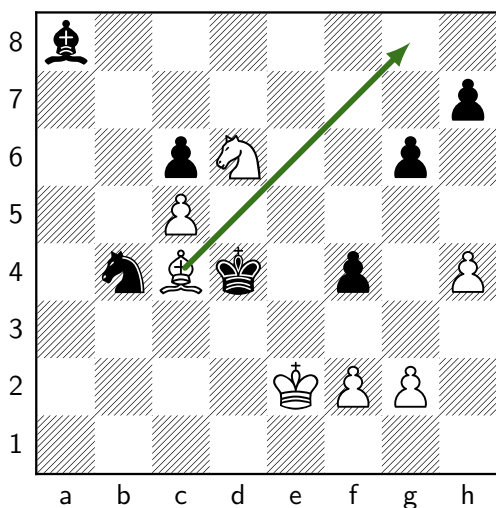
Figure 25: **Move 31, Black: Nf3+.** (a) Our position; Melanie plays Nf3+ (e5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#806` (ply 75) [structural distance $d=40$, QAP obj=52]: the master played Nd3+; under μ our Ne4 plays the role of that game’s Ne4; cross-type roles: our Ne5 as their rb7, our nc4 as their Bd7; transferred move e5f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3244` (ply 72) [structural distance $d=44$, QAP obj=30]: the master played Rb8; under μ our Ne4 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our Ba1 as their Nf1, our Bg4 as their Pf2; transferred move e5f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1525` (ply 73) [structural distance $d=41$, QAP obj=26]: the master played Ne5; under μ our Ne4 plays the role of that game’s Ng5; cross-type roles: our Pf7 as their Ne7, our Bg4 as their Pg6; transferred move e5f3 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Nf3+ (summed QAP objective 196, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



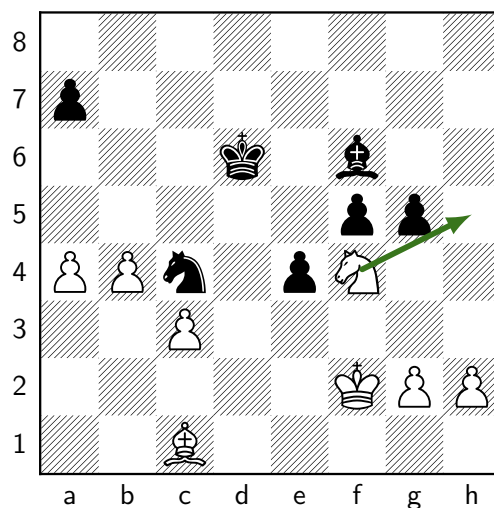
(a)



(b)

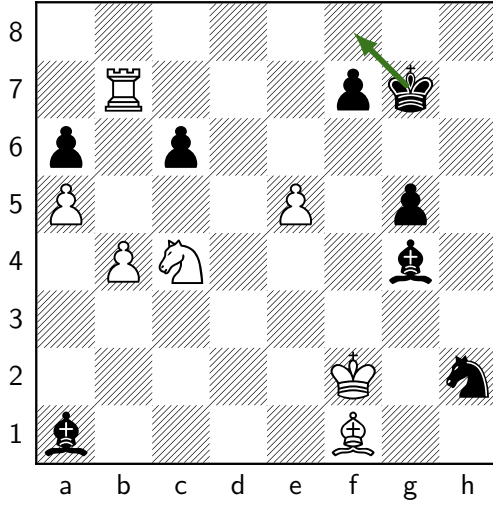


(c)

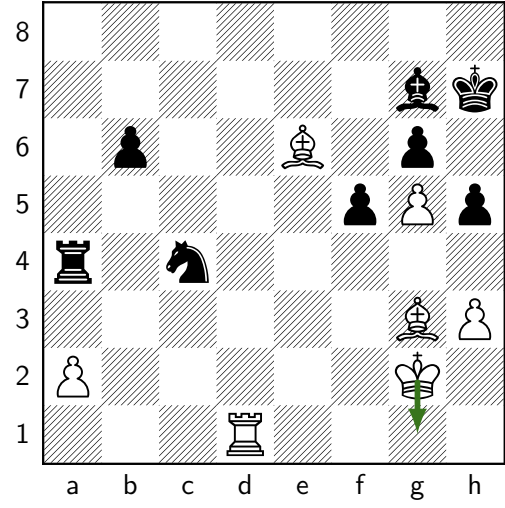


(d)

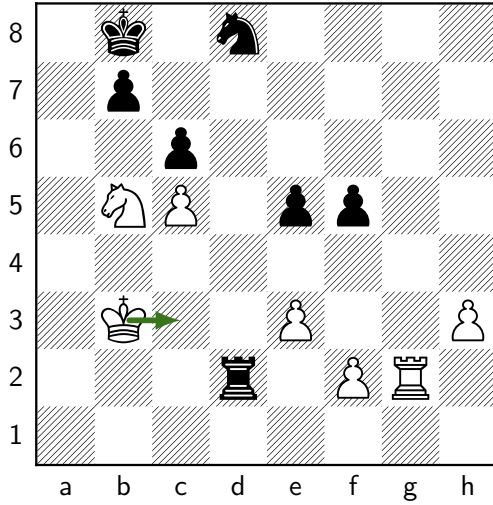
Figure 26: **Move 32, Black: Nh2**. (a) Our position; Melanie plays Nh2 (f3h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2307` (ply 76) [structural distance $d=38$, QAP obj=26]: the master played Nf8+; under μ our Nf6 plays the role of that game’s Ng6; cross-type roles: our bf1 as their ph6, our Pg5 as their rb8; transferred move f3h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2584` (ply 68) [structural distance $d=33$, QAP obj=25]: the master played Bg8; under μ our Nf6 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our rb7 as their Pg2, our Nf3 as their Pc5; transferred move f3h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#620` (ply 72) [structural distance $d=38$, QAP obj=18]: the master played Nh5; under μ our Nf6 plays the role of that game’s Nf4; cross-type roles: our rb7 as their Ph2, our bf1 as their Pa4; transferred move f3h2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh2 (summed QAP objective 69, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



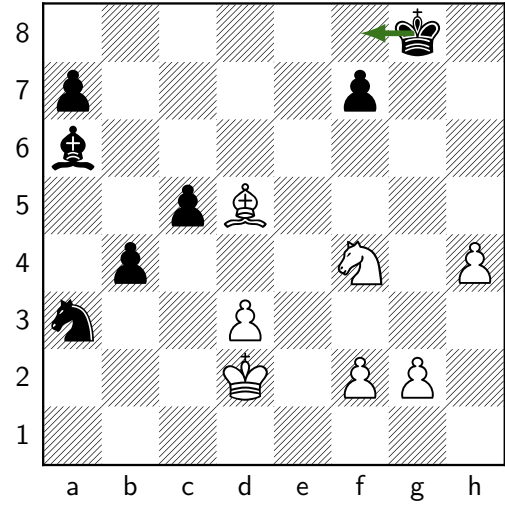
(a)



(b)

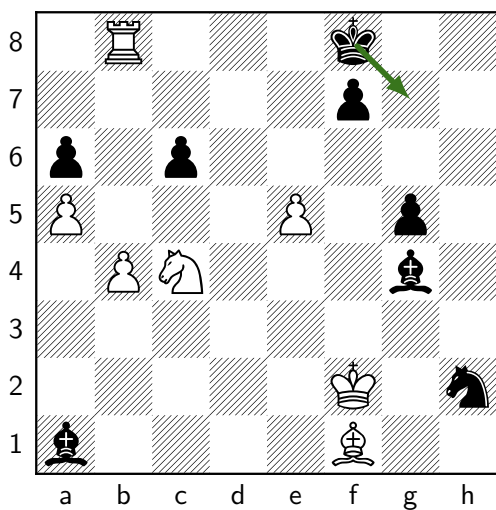


(c)

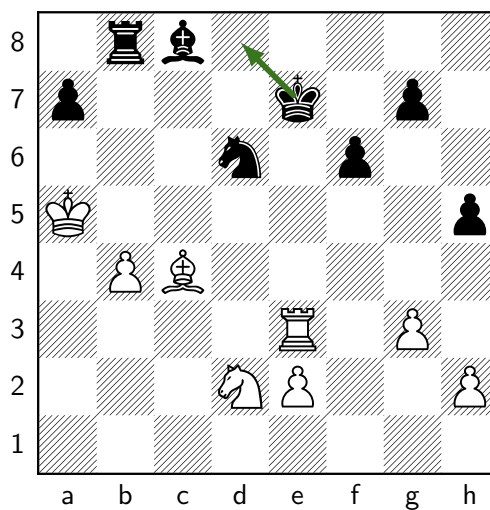


(d)

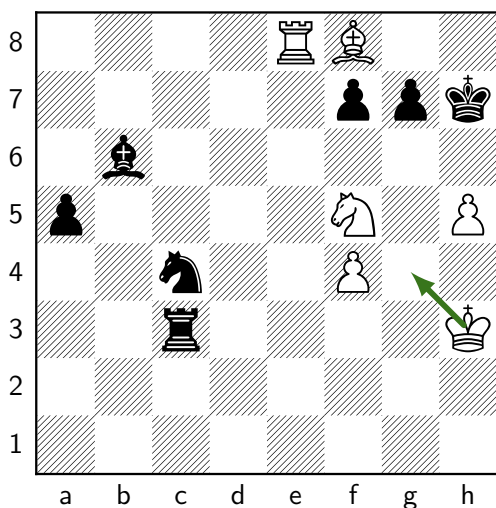
Figure 27: **Move 33, Black: Kf8 (g7f8).** (a) Our position; Melanie plays Kf8 (g7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1862` (ply 66) [structural distance $d=36$, QAP obj=30]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Bg4 as their Rd1, our Ba1 as their Pa2; transferred move g7f8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1684` (ply 76) [structural distance $d=44$, QAP obj=24]: the master played Kc3; under μ our Kg2 plays the role of that game’s Kb3; cross-type roles: our Pc6 as their Rg2, our Ba1 as their Ph3; transferred move g7f8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2336` (ply 59) [structural distance $d=39$, QAP obj=21]: the master played Kf8; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Nh2 as their Ba6, our rb7 as their bd5; transferred move g7f8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf8 (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



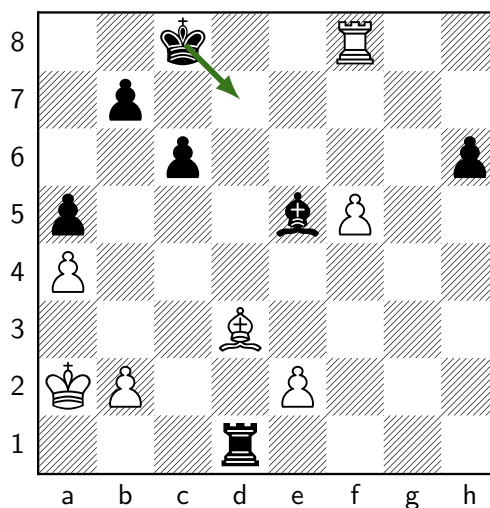
(a)



(b)

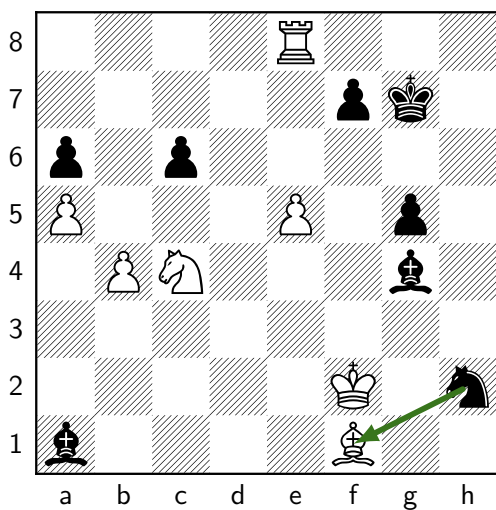


(c)

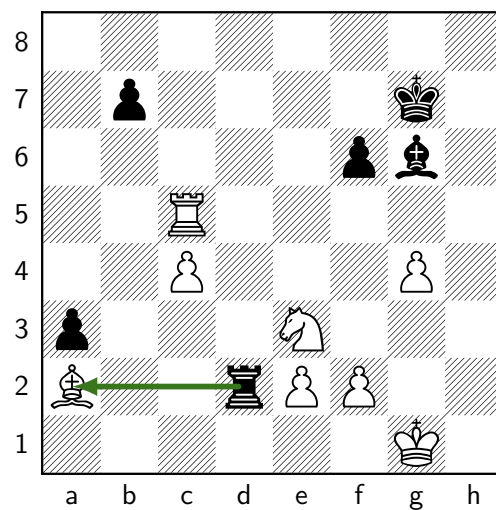


(d)

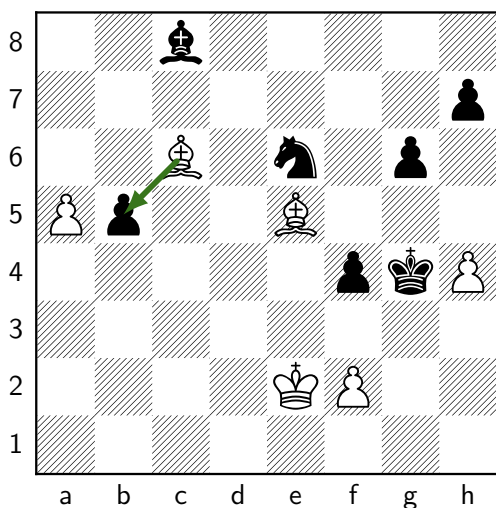
Figure 28: **Move 34, Black: Kg7**. (a) Our position; Melanie plays Kg7 (f8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1592` (ply 55) [structural distance $d=38$, QAP obj=69]: the master played Kd8; under μ our Kf1 plays the role of that game’s Ke2; cross-type roles: our Nh2 as their Rb8, our Ba1 as their Nd6; transferred move f8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2780` (ply 78) [structural distance $d=34$, QAP obj=57]: the master played Kg4; under μ our Kf1 plays the role of that game’s Kh3; cross-type roles: our bf1 as their nc4, our Bg4 as their Nf5; transferred move f8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2110` (ply 63) [structural distance $d=40$, QAP obj=54]: the master played Kd7; under μ our Kf1 plays the role of that game’s Kc1; cross-type roles: our Nh2 as their pb2, our nc4 as their pe2; transferred move f8g7 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kg7 (summed QAP objective 435, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



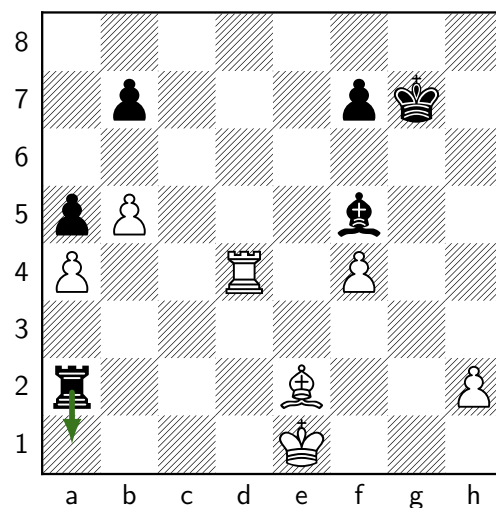
(a)



(b)

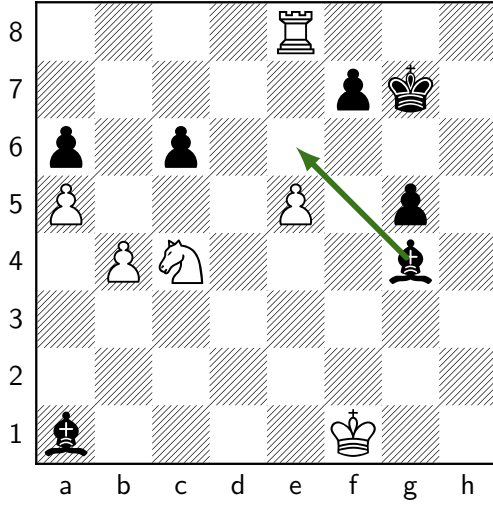


(c)

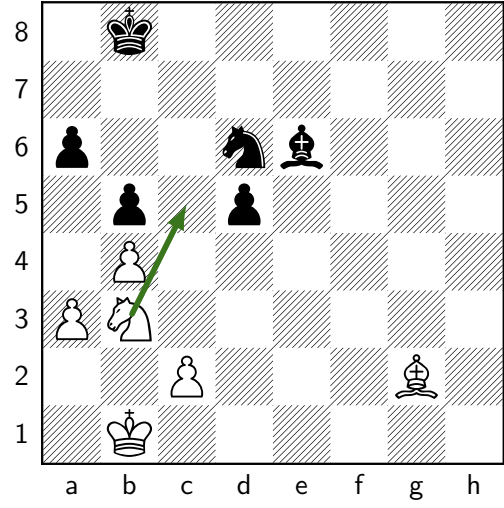


(d)

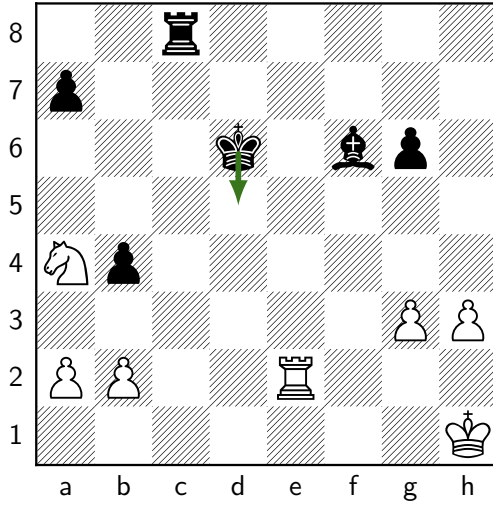
Figure 29: **Move 35, Black: Nxf1**. (a) Our position; Melanie plays Nxf1 (h2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2795` (ply 83) [structural distance $d=41$, QAP obj=22]: the master played Rxa2; under μ our Nh7 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Ba1 as their Pb7, our nc4 as their pc4; transferred move h2f1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2212` (ply 82) [structural distance $d=41$, QAP obj=18]: the master played Bxb5; under μ our Nh7 plays the role of that game’s Bc6 (a **cross-type** correspondence); cross-type roles: our bfl as their pf4, our re8 as their Pa5; transferred move h2f1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#489` (ply 77) [structural distance $d=40$, QAP obj=15]: the master played Ra1+; under μ our Nh7 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our Ba1 as their Pb7, our Bg4 as their Pa5; transferred move h2f1 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nxf1 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



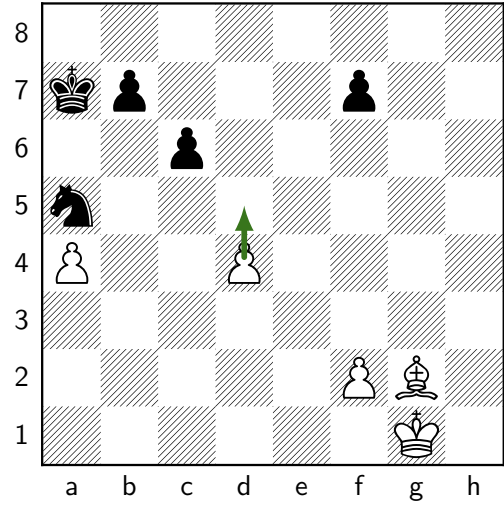
(a)



(b)

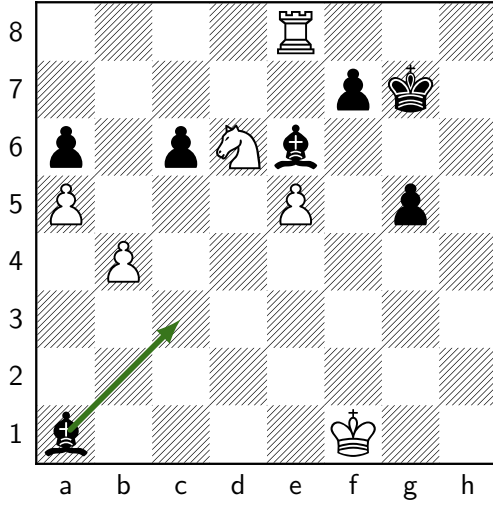


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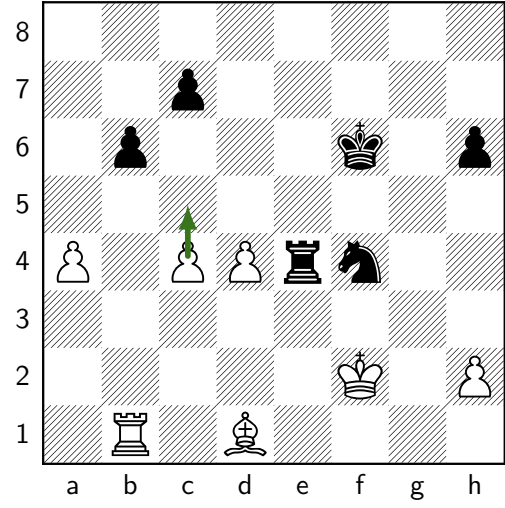


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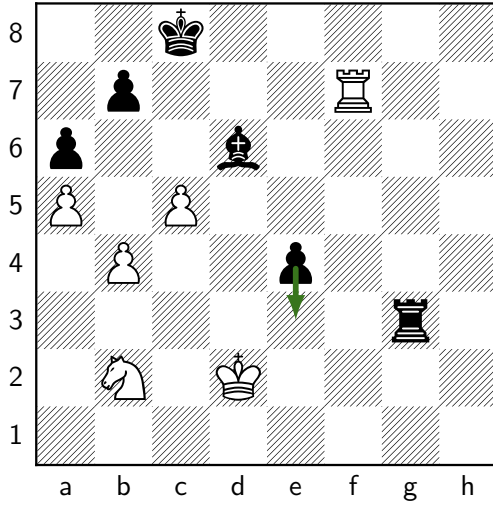
Figure 30: **Move 36, Black: Be6**. (a) Our position; Melanie plays **Be6** (**g4e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2971` (ply 72) [structural distance $d=28$, QAP obj=19]: the master played **Nc5**; under μ our **Bg5** plays the role of that game’s **Nb3** (a **cross-type** correspondence); cross-type roles: our **re8** as their **nd6**, our **nc4** as their **be6**; transferred move **g4e6** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3614` (ply 67) [structural distance $d=26$, QAP obj=23]: the master played **Kd5**; under μ our **Kg2** plays the role of that game’s **Kd3**; cross-type roles: our **Pf7** as their **Rc8**, our **Bg4** as their **ph3**; transferred move **g7h7**. (d) Retrieved from `ccr1_top_p0000#1618` (ply 78) [structural distance $d=27$, QAP obj=22]: the master played **d5**; under μ our **Bg5** plays the role of that game’s **Pd4** (a **cross-type** correspondence); cross-type roles: our **re8** as their **pb7**; transferred move **g4f3**. Synthesis: 1 retrieved analog transferred to **Be6** (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



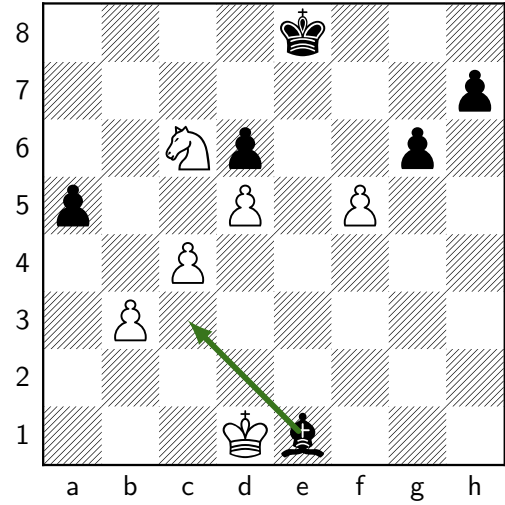
(a)



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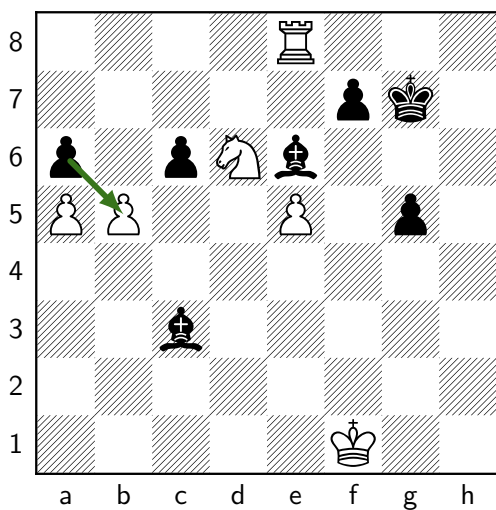


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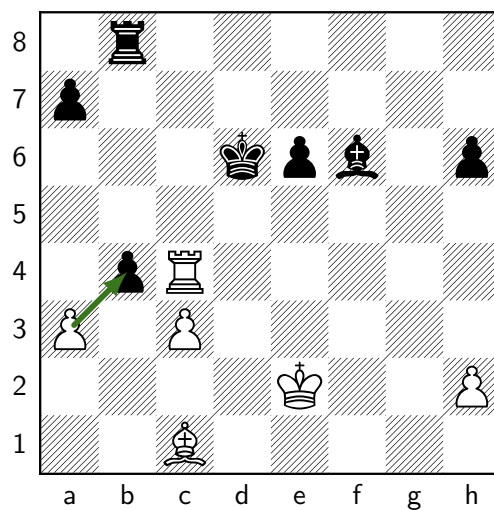


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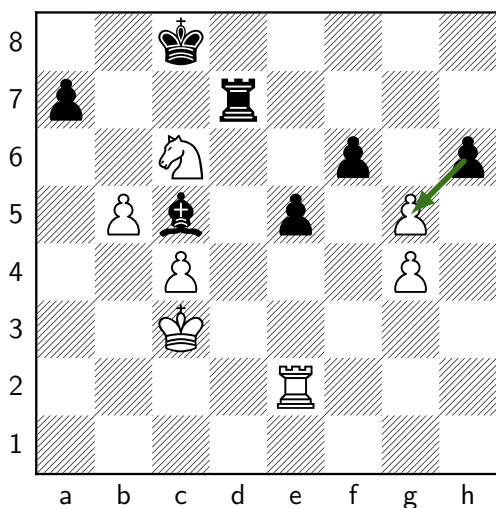
Figure 31: **Move 37, Black: Bc3**. (a) Our position; Melanie plays Bc3 (a1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#186` (ply 74) [structural distance $d=24$, QAP obj=16]: the master played c5; under μ our *Ba8* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *pa5* as their *Rb1*, our *nd6* as their *Bd1*; transferred move a1c3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2220` (ply 81) [structural distance $d=21$, QAP obj=14]: the master played e3+; under μ our *Ba8* plays the role of that game’s *Pe5* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *rf7*, our *re8* as their *pc5*; transferred move a1c3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3506` (ply 77) [structural distance $d=22$, QAP obj=11]: the master played Bc3; under μ our *Ba8* plays the role of that game’s *Be8*; cross-type roles: our *pe5* as their *nc6*, our *Be6* as their *Pg6*; transferred move a1c3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc3 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



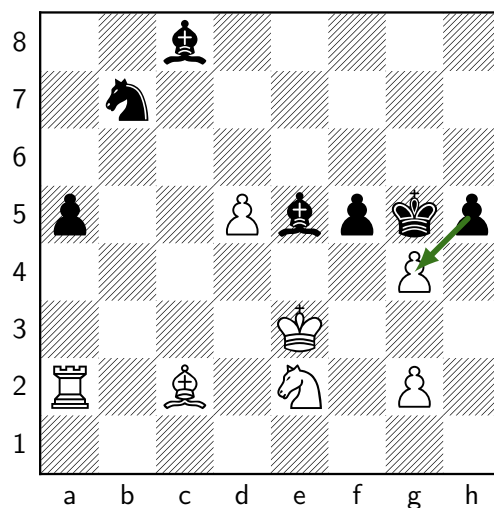
(a)



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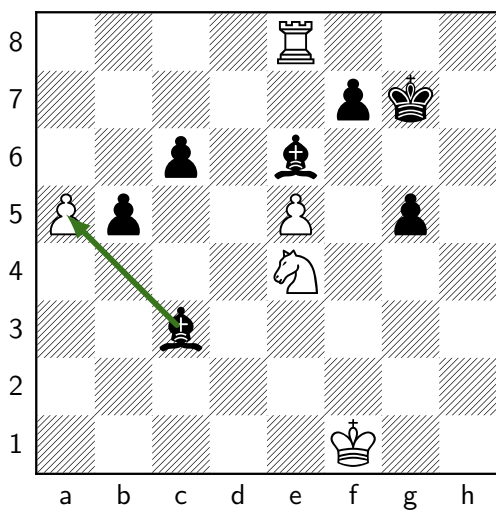


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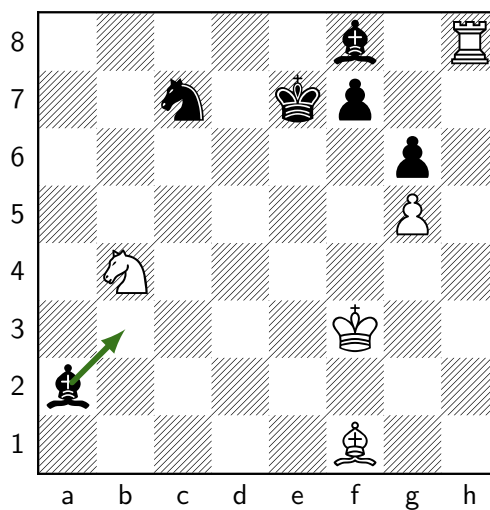


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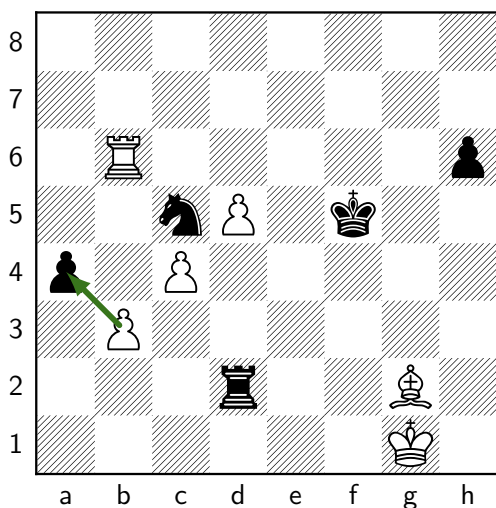
Figure 32: **Move 38, Black: axb5.** (a) Our position; Melanie plays **axb5** (**a6b5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0001#2587` (ply 70) [structural distance $d=51$, QAP obj=84]: the master played **axb4**; under μ our *Pa3* plays the role of that game’s *Pa3*; cross-type roles: our *pe5* as their *Bc1*, our *re8* as their *Ph2*; transferred move **a6b5** (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#745` (ply 87) [structural distance $d=51$, QAP obj=80]: the master played **hxc5**; under μ our *Pa3* plays the role of that game’s *Ph3*; cross-type roles: our *Bc3* as their *Rd7*, our *re8* as their *pc4*; transferred move **a6b5** (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#3413` (ply 81) [structural distance $d=56$, QAP obj=70]: the master played **hxc4**; under μ our *Pa3* plays the role of that game’s *Ph4*; cross-type roles: our *nd6* as their *Bc8*, our *re8* as their *Nb7*; transferred move **a6b5** (**endorses this move**). Synthesis: 13 retrieved analogs transferred to **axb5** (summed QAP objective 567, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



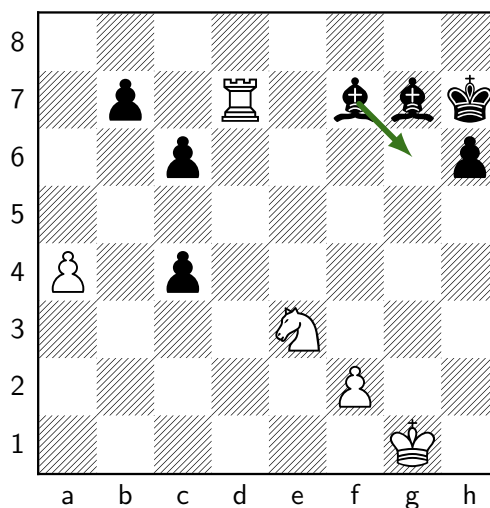
(a)



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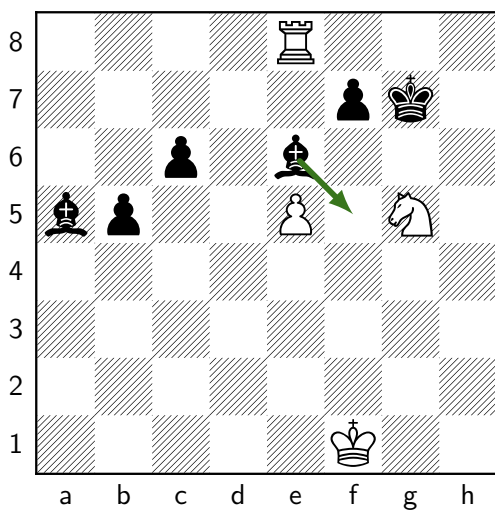


(c)

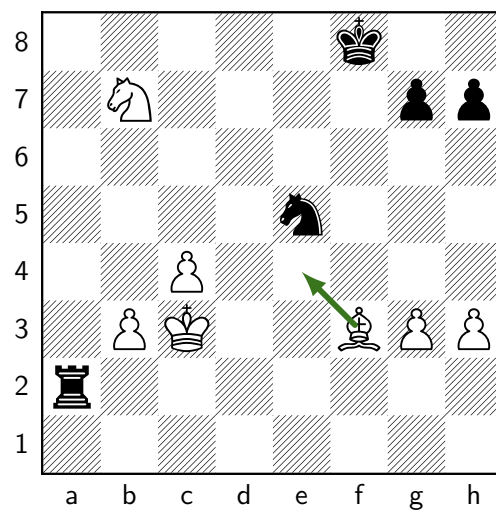


(d)

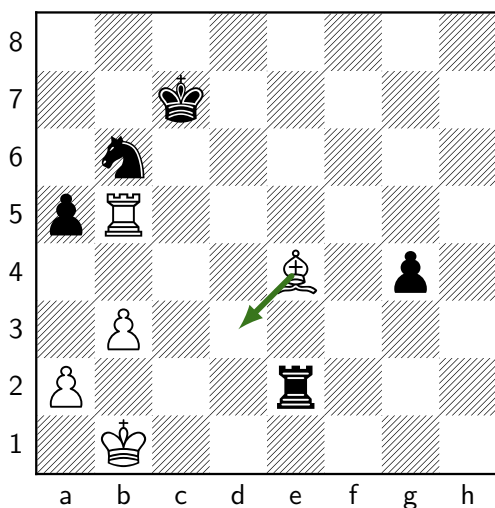
Figure 33: **Move 39, Black: Bxa5.** (a) Our position; Melanie plays Bxa5 (c3a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1629` (ply 91) [structural distance $d=36$, QAP obj=41]: the master played Bb3; under μ our Bc6 plays the role of that game’s Ba7; cross-type roles: our Pc6 as their Nc7, our Pb5 as their bf1; transferred move c3a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2385` (ply 82) [structural distance $d=33$, QAP obj=36]: the master played bxa4; under μ our Bc6 plays the role of that game’s Pb3 (a **cross-type** correspondence); cross-type roles: our Pg5 as their Rb6; transferred move c3a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2016` (ply 87) [structural distance $d=32$, QAP obj=35]: the master played Bg6; under μ our Bc6 plays the role of that game’s Bf2; cross-type roles: our ne4 as their rd7, our Pc6 as their Bg7; transferred move c3a5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bxa5 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



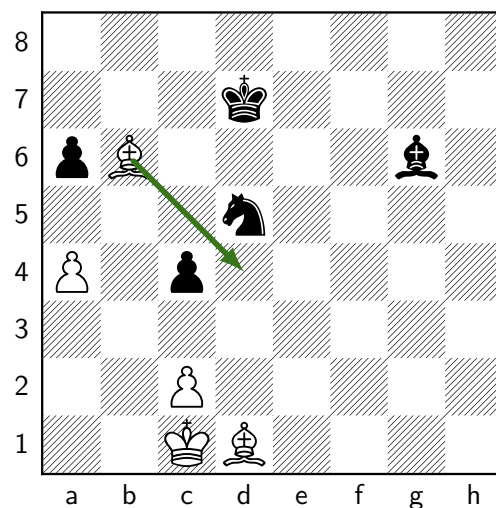
(a)



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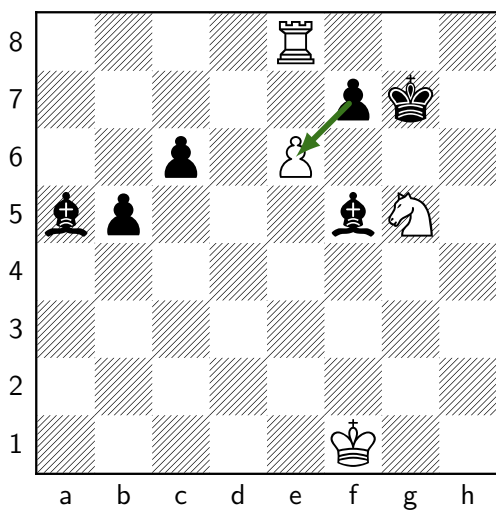


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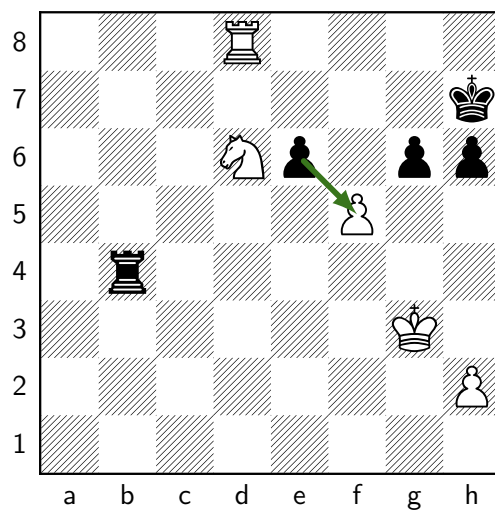


(d)

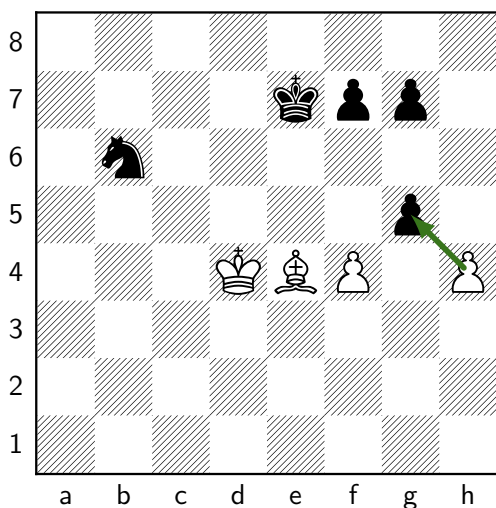
Figure 34: **Move 40, Black: Bf5 (e6f5)**. (a) Our position; Melanie plays Bf5 (e6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1477` (ply 92) [structural distance $d=25$, QAP obj=31]: the master played Be4; under μ our Be3 plays the role of that game’s Bf3; cross-type roles: our Ba5 as their Nb7; transferred move e6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3459` (ply 86) [structural distance $d=28$, QAP obj=27]: the master played Bd3; under μ our Be3 plays the role of that game’s Be4; cross-type roles: our ng5 as their re2, our Ba5 as their pg4; transferred move e6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1766` (ply 90) [structural distance $d=22$, QAP obj=21]: the master played Bd4; under μ our Be3 plays the role of that game’s Bb6; cross-type roles: our re8 as their nd5, our ng5 as their bg6; transferred move e6f5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bf5 (summed QAP objective 139, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



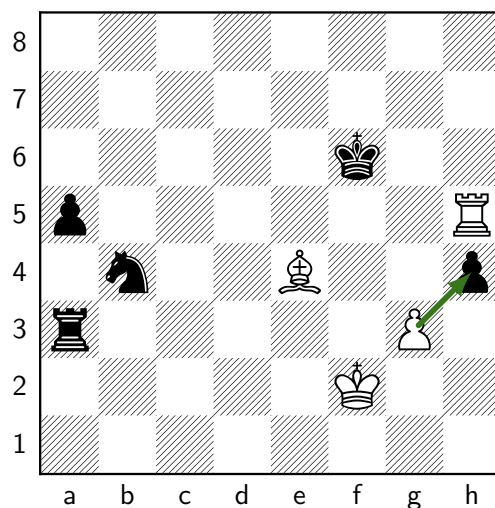
(a)



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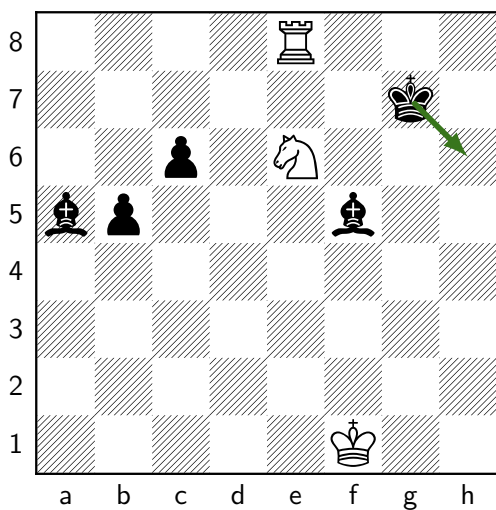


(c)

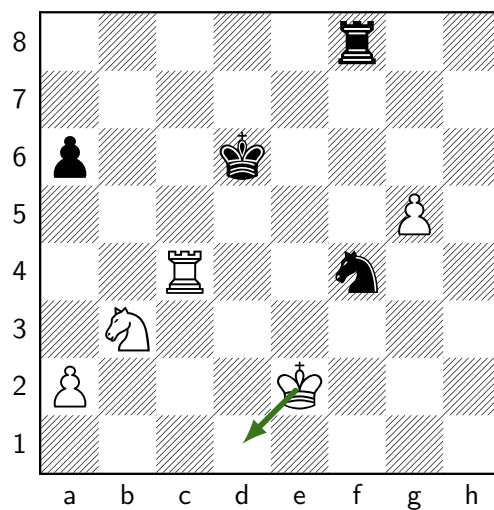


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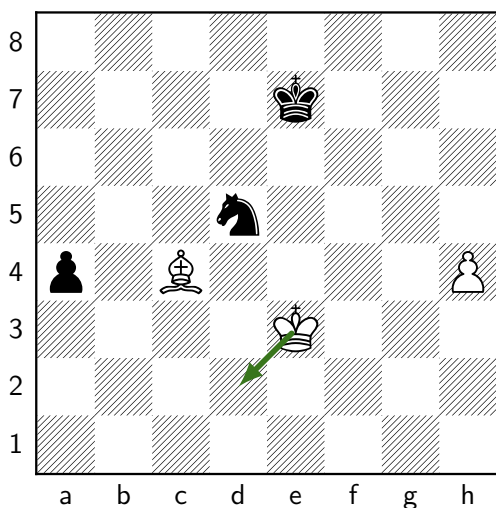
Figure 35: **Move 41, Black: fxe6**. (a) Our position; Melanie plays **fxe6** (**f7e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2144` (ply 83) [structural distance $d=28$, QAP obj=75]: the master played **exf5**; under μ our *Pf2* plays the role of that game’s *Pe3*; cross-type roles: our *ng5* as their *rd8*, our *re8* as their *nd6*; transferred move **f7e6** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1444` (ply 86) [structural distance $d=33$, QAP obj=69]: the master played **hxxg5**; under μ our *Pf2* plays the role of that game’s *Ph4*; cross-type roles: our *re8* as their *pg7*; transferred move **f7e6** (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1569` (ply 78) [structural distance $d=31$, QAP obj=69]: the master played **gxh4**; under μ our *Pf2* plays the role of that game’s *Pg3*; cross-type roles: our *ng5* as their *ra3*, our *Pb5* as their *nb4*; transferred move **f7e6** (**endorses this move**). Synthesis: 14 retrieved analogs transferred to **fxe6** (summed QAP objective 687, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



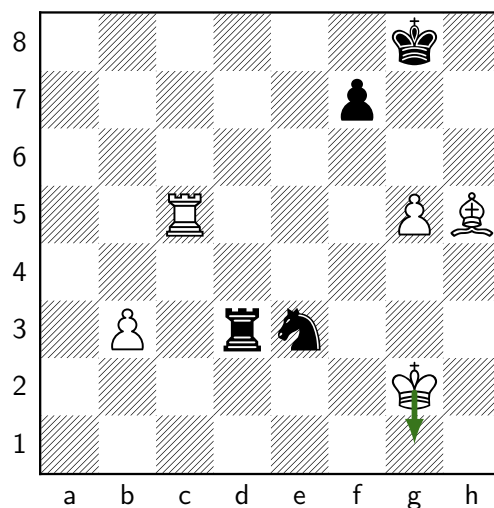
(a)



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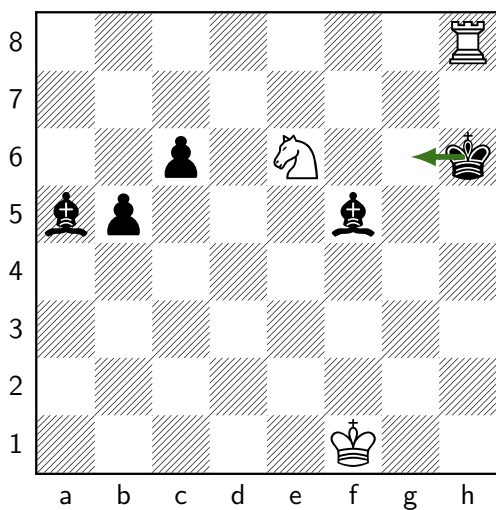


(c)

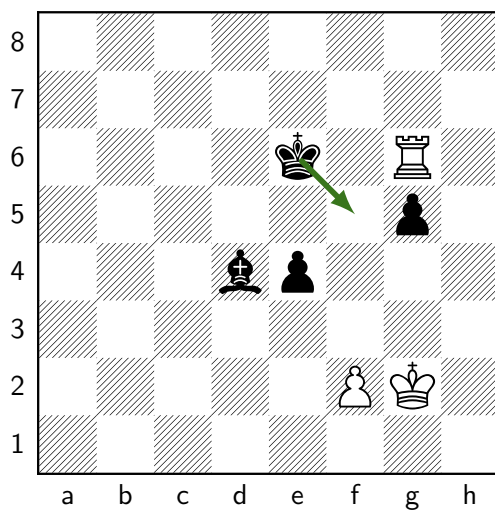


(d)

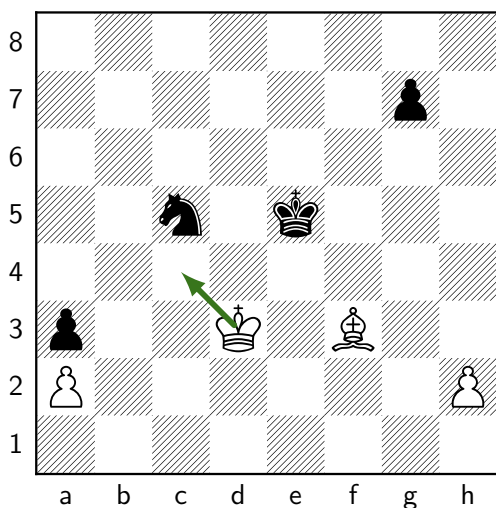
Figure 36: **Move 42, Black: Kh6 (g7h6)**. (a) Our position; Melanie plays Kh6 (g7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2007` (ply 86) [structural distance $d=16$, QAP obj=74]: the master played Kd1; under μ our Kg2 plays the role of that game’s Ke2; cross-type roles: our Pb5 as their Nb3, our Bf5 as their Rc4; transferred move g7h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2070` (ply 98) [structural distance $d=21$, QAP obj=71]: the master played Kd2; under μ our Kg2 plays the role of that game’s Ke3; cross-type roles: our re8 as their pa4; transferred move g7h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1018` (ply 92) [structural distance $d=18$, QAP obj=69]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Bf5 as their Rc5; transferred move g7h6 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kh6 (summed QAP objective 479, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



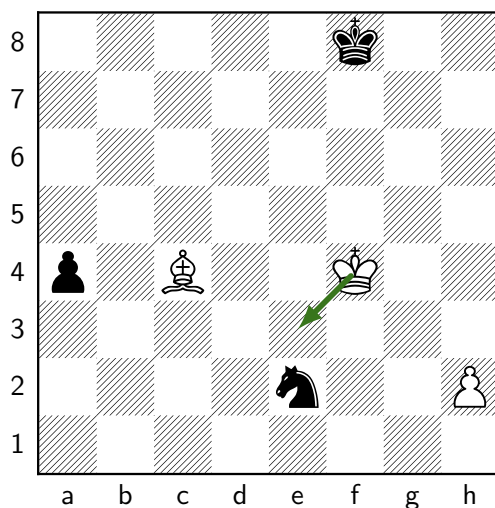
(a)



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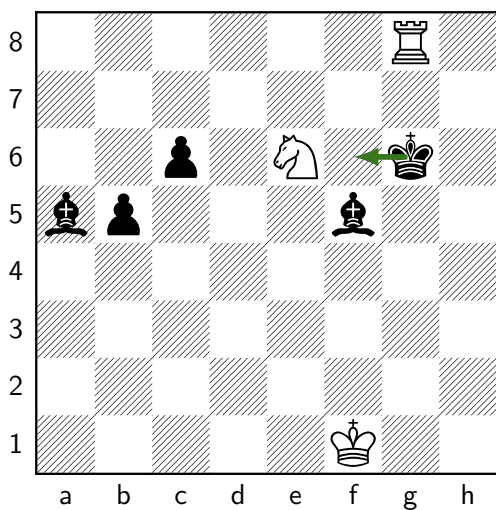


(c)

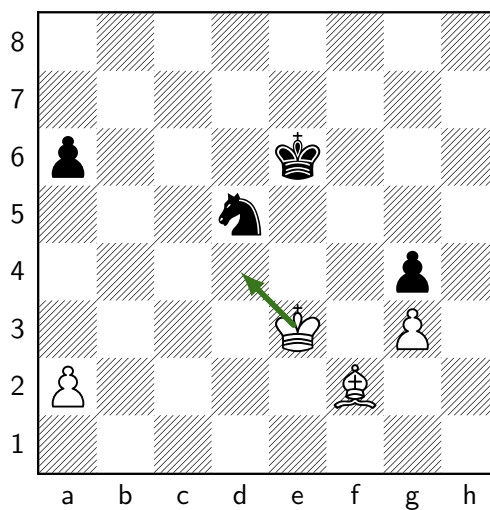


(d)

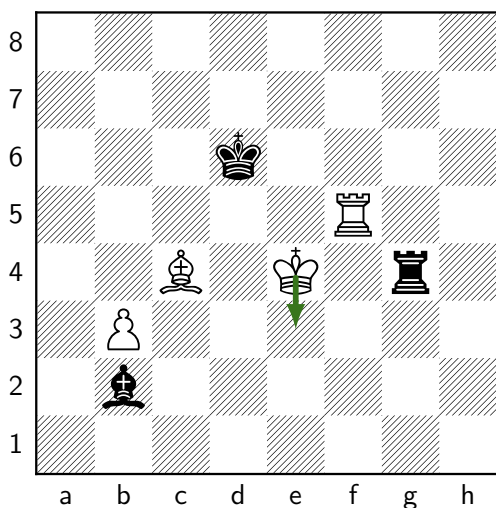
Figure 37: **Move 43, Black: Kg6**. (a) Our position; Melanie plays Kg6 (**h6g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2654` (ply 95) [structural distance $d=16$, QAP obj=67]: the master played Kf5; under μ our Kh3 plays the role of that game’s Ke3; cross-type roles: our Ba5 as their Pe4, our ne6 as their pf2; transferred move **h6g6 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#2948` (ply 100) [structural distance $d=15$, QAP obj=67]: the master played Kc4; under μ our Kh3 plays the role of that game’s Kd3; cross-type roles: our Ba5 as their Pa2, our ne6 as their pa3; transferred move **h6g6 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#2070` (ply 88) [structural distance $d=13$, QAP obj=67]: the master played Ke3; under μ our Kh3 plays the role of that game’s Kf4; cross-type roles: our rh8 as their ne2, our Bf5 as their Ph2; transferred move **h6g6 (endorses this move)**. Synthesis: 20 retrieved analogs transferred to Kg6 (summed QAP objective 583, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



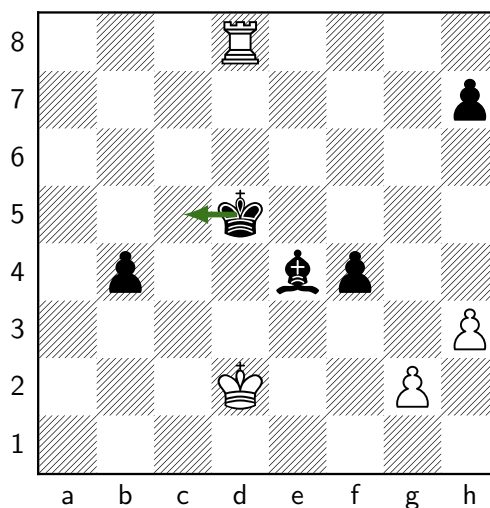
(a)



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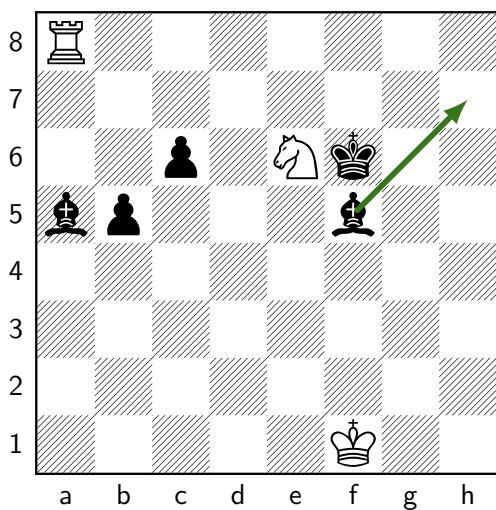


(c)

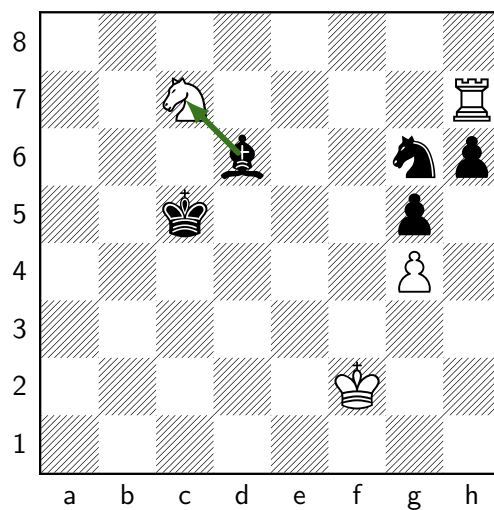


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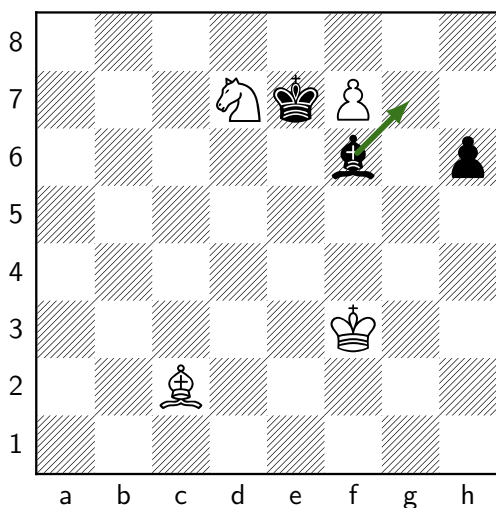
Figure 38: **Move 44, Black: Kf6**. (a) Our position; Melanie plays Kf6 (g6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2369 (ply 90) [structural distance $d=21$, QAP obj=78]: the master played Kd4; under μ our Kg3 plays the role of that game’s Ke3; cross-type roles: our Ba5 as their Pg3, our ne6 as their pg4; transferred move g6f6 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#164 (ply 92) [structural distance $d=19$, QAP obj=70]: the master played Ke3; under μ our Kg3 plays the role of that game’s Ke4; cross-type roles: our ne6 as their bb2, our Bf5 as their Rf5; transferred move g6f6 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1191 (ply 101) [structural distance $d=20$, QAP obj=54]: the master played Kc5; under μ our Kg3 plays the role of that game’s Kd4; cross-type roles: our Ba5 as their Pb4, our ne6 as their Pf4; transferred move g6f6 (**endorses this move**). Synthesis: 16 retrieved analogs transferred to Kf6 (summed QAP objective 576, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



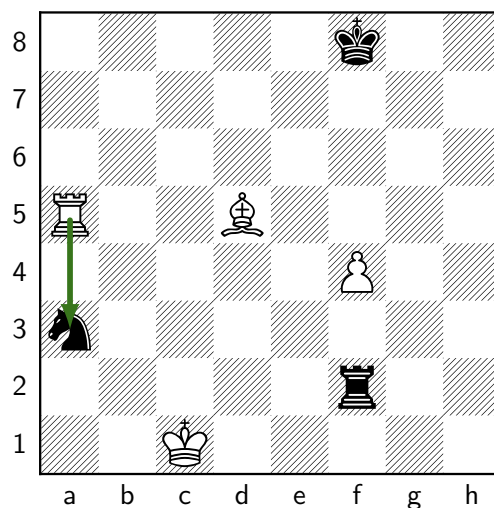
(a)



(b)

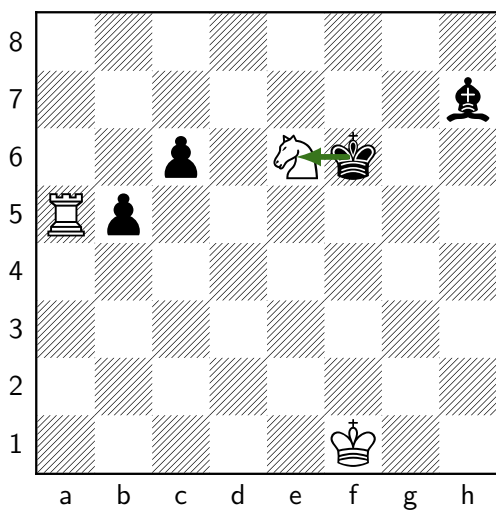


(c)

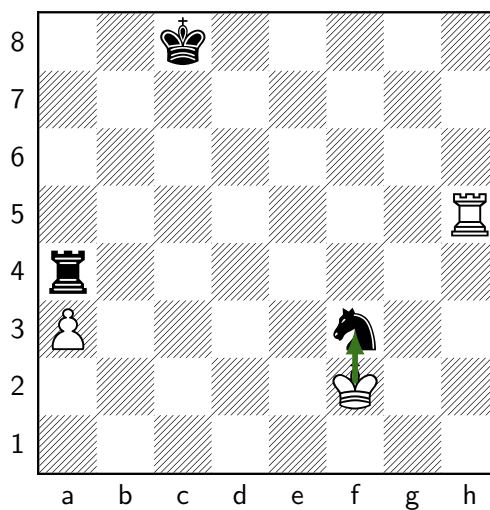


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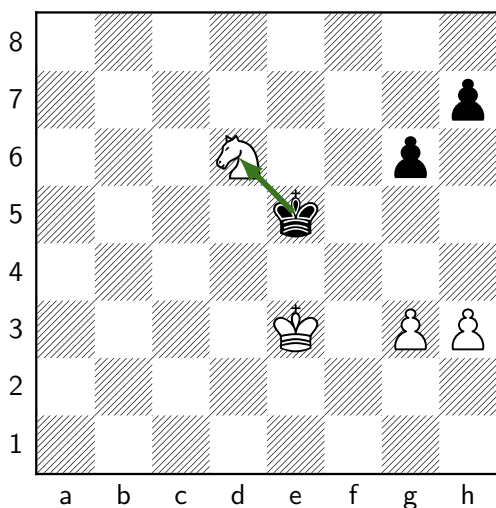
Figure 39: **Move 45, Black: Bh7 (f5h7)**. (a) Our position; Melanie plays Bh7 (f5h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1209` (ply 105) [structural distance $d=21$, QAP obj=26]: the master played Bxc7; under μ our Bf4 plays the role of that game’s Bd3; cross-type roles: our Ba5 as their Ph6; transferred move f5h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2986` (ply 97) [structural distance $d=23$, QAP obj=21]: the master played Bg7; under μ our Bf4 plays the role of that game’s Bf3; cross-type roles: our ne6 as their pf7, our ra8 as their Ph6; transferred move f5h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2165` (ply 100) [structural distance $d=19$, QAP obj=10]: the master played Rxa3; under μ our Bf4 plays the role of that game’s Ra5 (a **cross-type** correspondence); transferred move f5h7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bh7 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



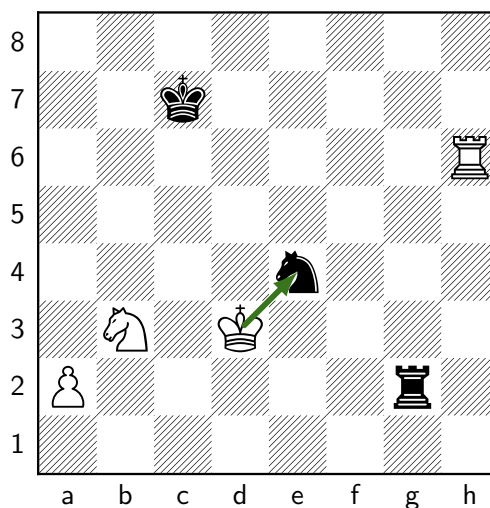
(a)



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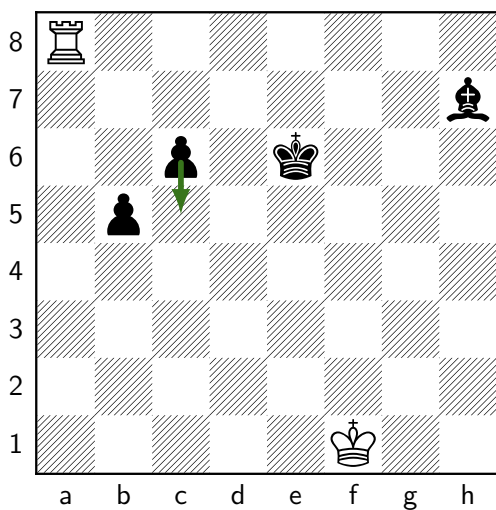


(c)

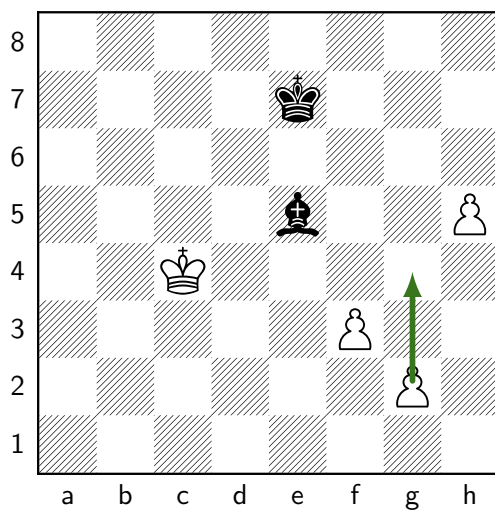


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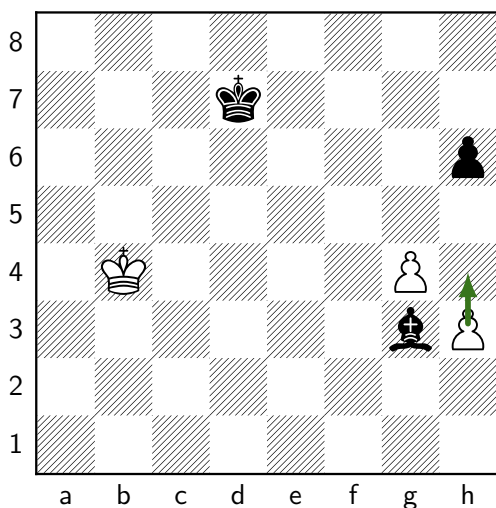
Figure 40: **Move 46, Black: Kxe6**. (a) Our position; Melanie plays Kxe6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3211` (ply 106) [structural distance $d=8$, QAP obj=25]: the master played Kxf3; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Pc6 as their Rh5; transferred move f6e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#134` (ply 79) [structural distance $d=11$, QAP obj=20]: the master played Kxd6; under μ our Kf3 plays the role of that game’s Ke4; cross-type roles: our ra5 as their pg3, our Bh7 as their ph3; transferred move f6e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2007` (ply 102) [structural distance $d=9$, QAP obj=20]: the master played Kxe4; under μ our Kf3 plays the role of that game’s Kd3; cross-type roles: our Bh7 as their Pa2, our Pc6 as their Nb3; transferred move f6e6 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Kxe6 (summed QAP objective 101, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



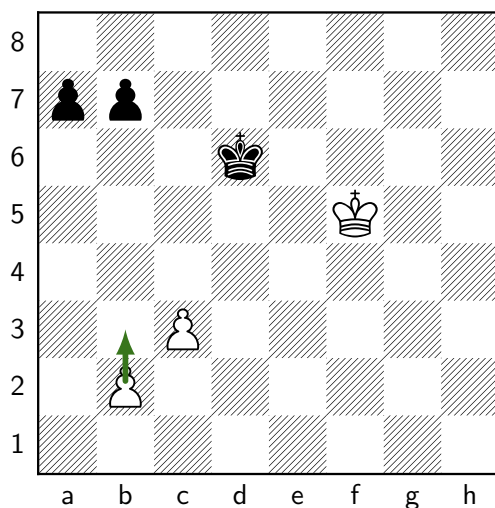
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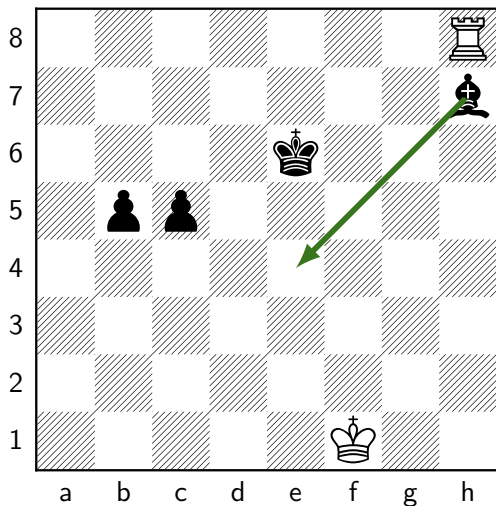


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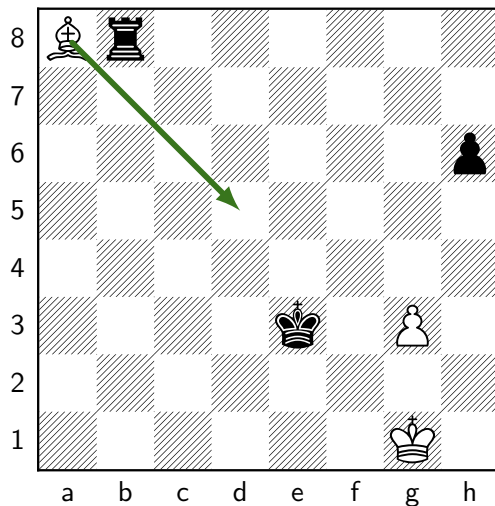


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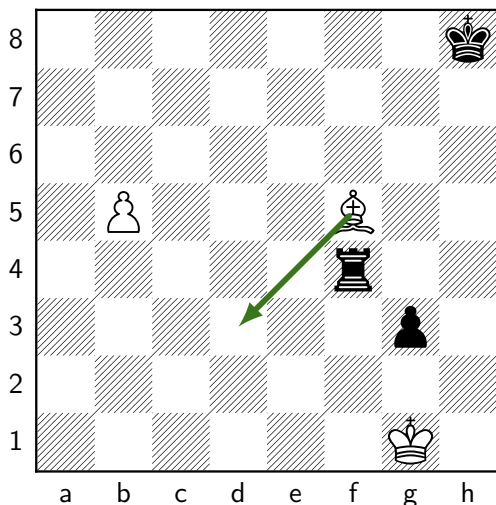
Figure 41: **Move 47, Black: c5**. (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2424` (ply 104) [structural distance $d=5$, QAP obj=2]: the master played g4; under μ our *Pc3* plays the role of that game’s *Pg2*; cross-type roles: our *ra8* as their *be5*, our *Bh7* as their *Ph5*; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2716` (ply 100) [structural distance $d=6$, QAP obj=1]: the master played h4; under μ our *Pc3* plays the role of that game’s *Ph3*; cross-type roles: our *ra8* as their *ph6*; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#721` (ply 106) [structural distance $d=7$, QAP obj=1]: the master played b3; under μ our *Pc3* plays the role of that game’s *Pb2*; cross-type roles: our *Bh7* as their *pa7*, our *ra8* as their *pb7*; transferred move c6c5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to c5 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



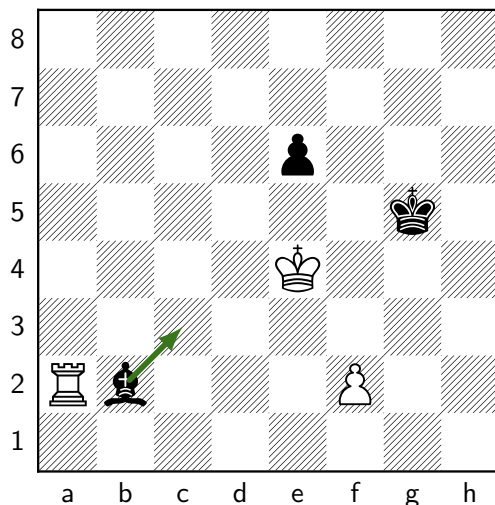
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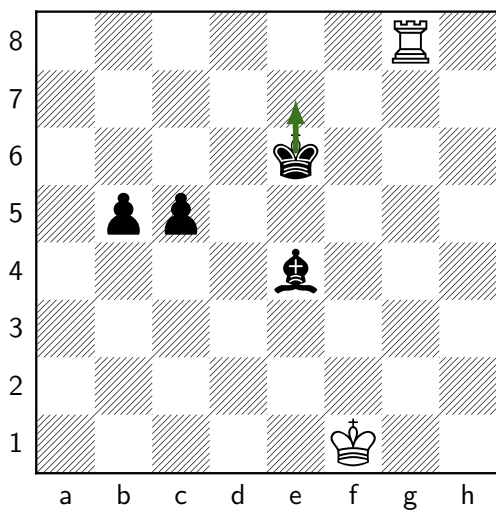


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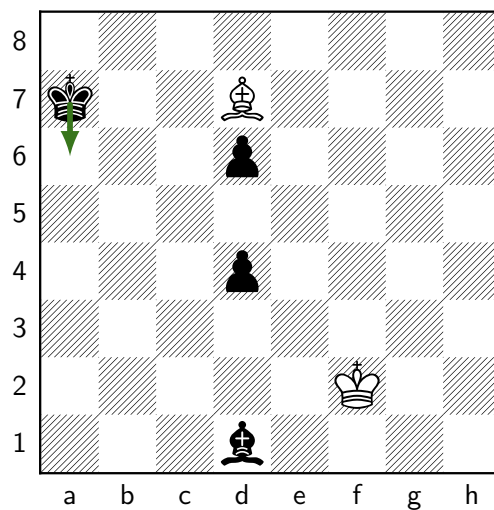


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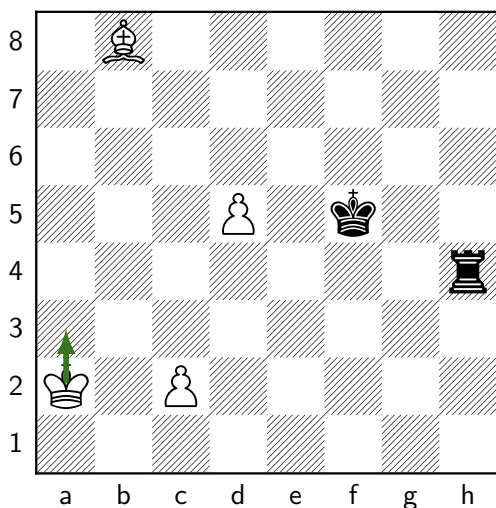
Figure 42: **Move 48, Black: Be4**. (a) Our position; Melanie plays **Be4** (**h7e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3664` (ply 106) [structural distance $d=3$, QAP obj=18]: the master played **Bd5**; under μ our *Bh2* plays the role of that game’s *Ba8*; transferred move **h7e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2531` (ply 94) [structural distance $d=6$, QAP obj=18]: the master played **Bd3**; under μ our *Bh2* plays the role of that game’s *Bf5*; transferred move **h7e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#994` (ply 101) [structural distance $d=6$, QAP obj=18]: the master played **Bc3**; under μ our *Bh2* plays the role of that game’s *Bb7*; transferred move **h7e4** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **Be4** (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



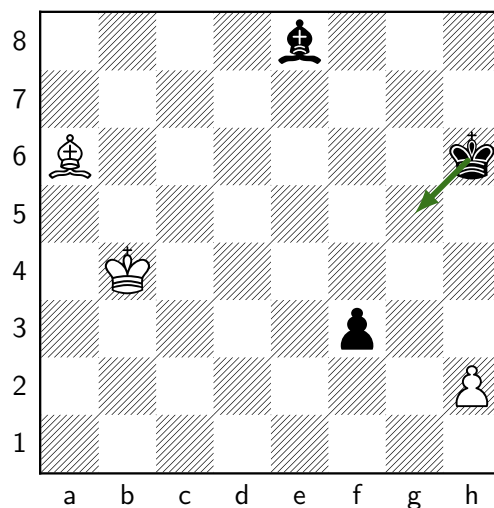
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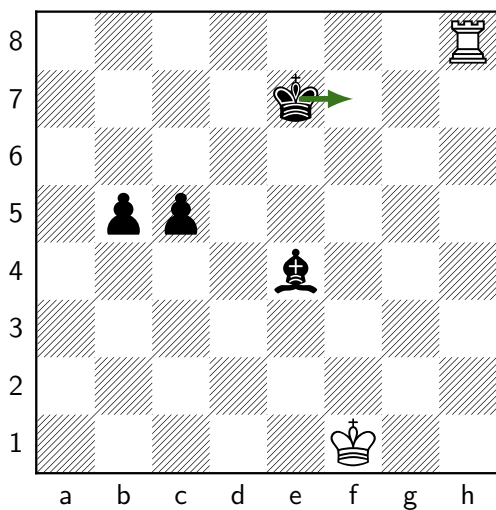


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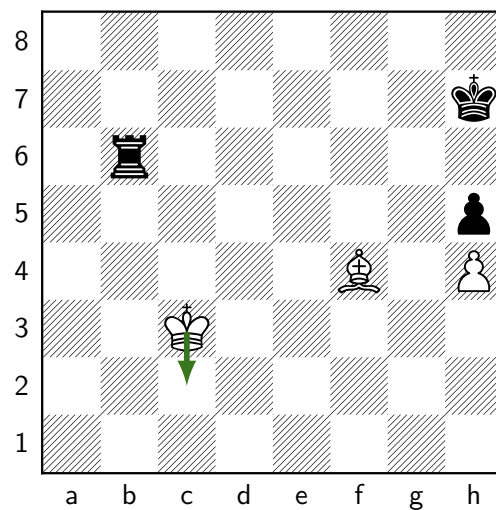


(d)

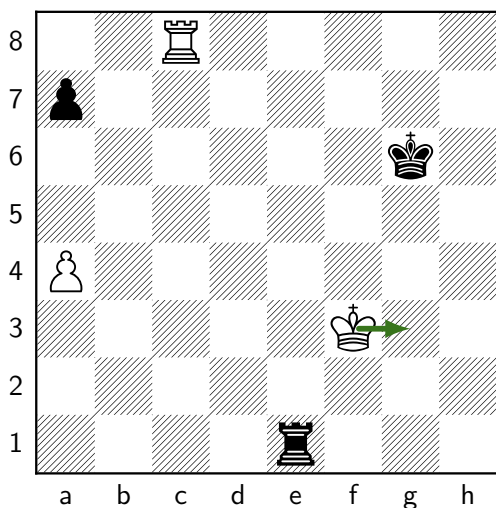
Figure 43: **Move 49, Black: Ke7.** (a) Our position; Melanie plays Ke7 (e6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2609 (ply 101) [structural distance $d=3$, QAP obj=1]: the master played Ka6; under μ our Ke3 plays the role of that game’s Ka2; cross-type roles: our rg8 as their bd7; transferred move e6e7 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3159 (ply 92) [structural distance $d=1$, QAP obj=1]: the master played Ka3; under μ our Ke3 plays the role of that game’s Ka2; transferred move e6e7 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3988 (ply 111) [structural distance $d=5$, QAP obj=1]: the master played Kg5; under μ our Ke3 plays the role of that game’s Kh3; cross-type roles: our rg8 as their ba6; transferred move e6e7 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Ke7 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



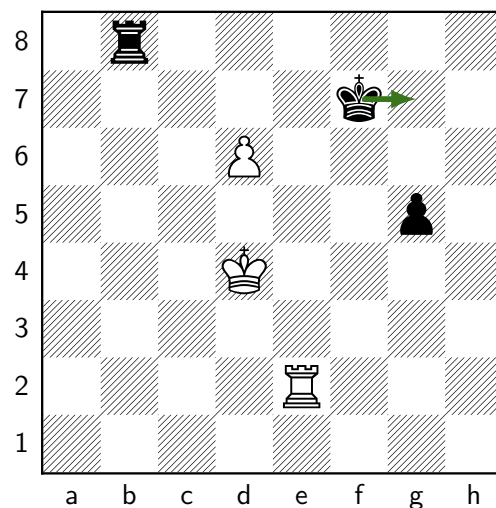
(a)



(b)

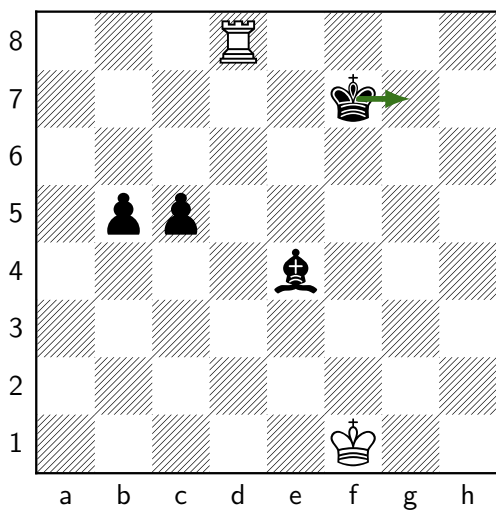


(c)

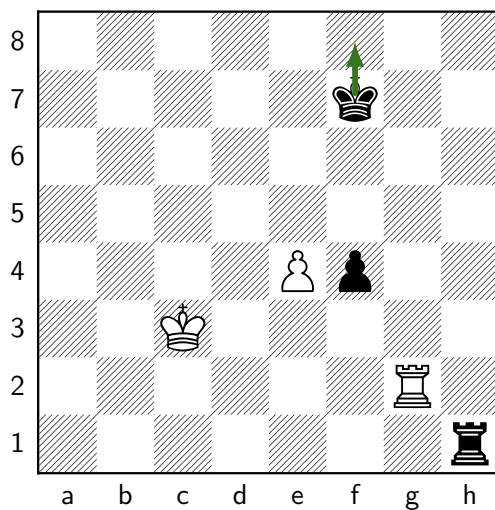


(d)

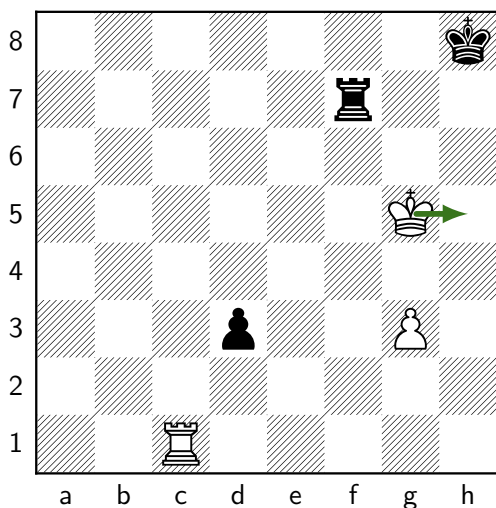
Figure 44: **Move 50, Black: Kf7.** (a) Our position; Melanie plays Kf7 (e7f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3532 (ply 100) [structural distance $d=3$, QAP obj=1]: the master played Kc2; under μ our Ke2 plays the role of that game’s Kc3; transferred move e7f7 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#1021 (ply 98) [structural distance $d=4$, QAP obj=1]: the master played Kg3; under μ our Ke2 plays the role of that game’s Kf3; cross-type roles: our Be4 as their Rc8; transferred move e7f7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#329 (ply 97) [structural distance $d=5$, QAP obj=1]: the master played Kg7; under μ our Ke2 plays the role of that game’s Kf2; cross-type roles: our Be4 as their Rb8; transferred move e7f7 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kf7 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



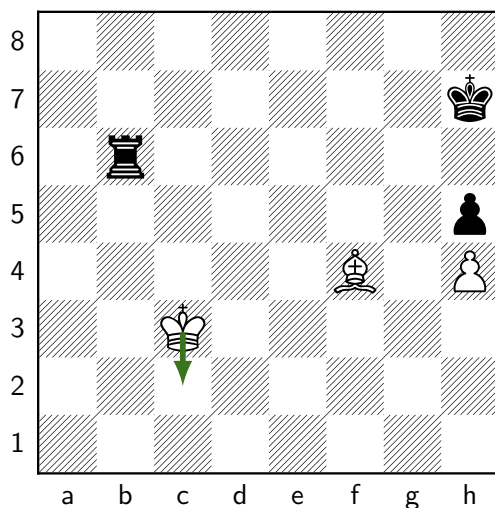
(a)



(b)

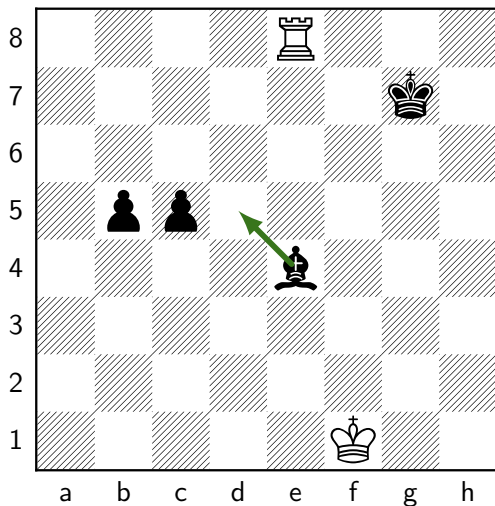


(c)

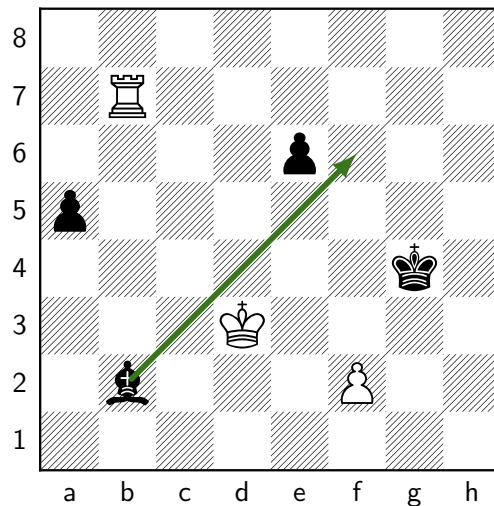


(d)

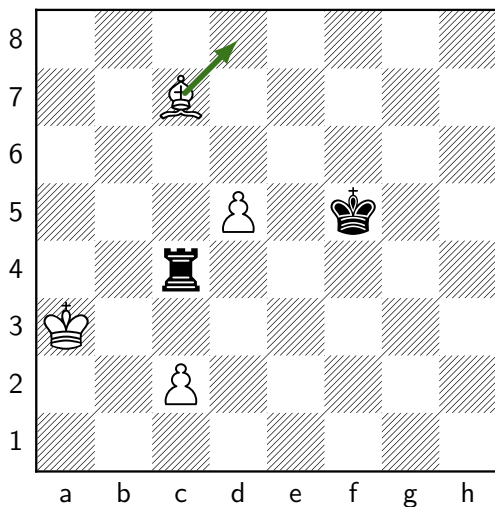
Figure 45: **Move 51, Black: Kg7.** (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2286` (ply 117) [structural distance $d=4$, QAP obj=2]: the master played Kf8; under μ our Kf2 plays the role of that game’s Kf2; cross-type roles: our Be4 as their Rh1; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3265` (ply 108) [structural distance $d=5$, QAP obj=2]: the master played Kh5; under μ our Kf2 plays the role of that game’s Kg5; cross-type roles: our Be4 as their Rc1; transferred move f7g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3532` (ply 100) [structural distance $d=3$, QAP obj=1]: the master played Kc2; under μ our Kf2 plays the role of that game’s Kc3; transferred move f7g7 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Kg7 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



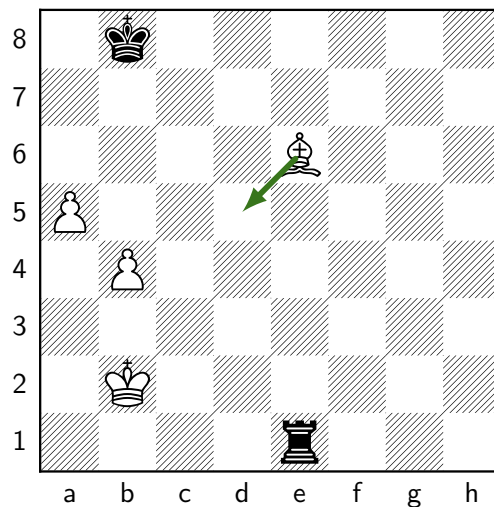
(a)



(b)

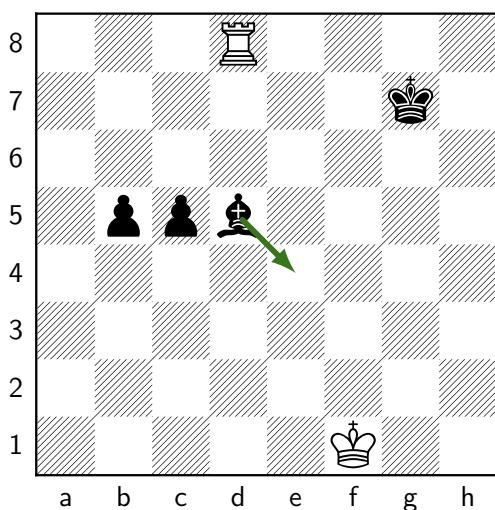


(c)

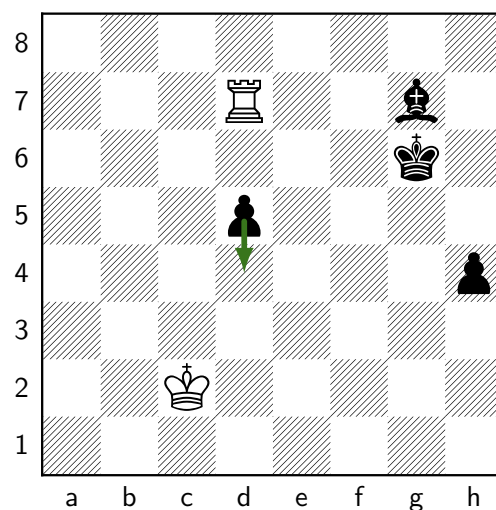


(d)

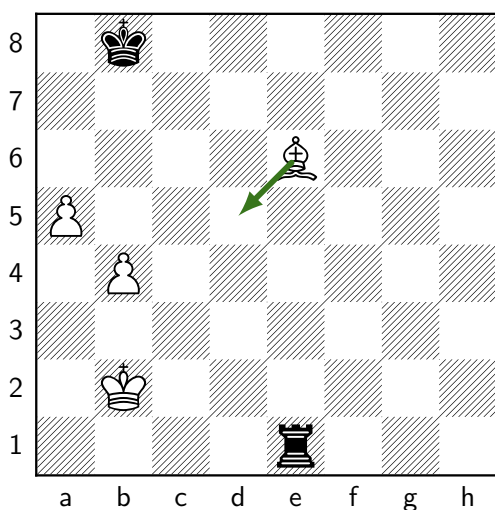
Figure 46: **Move 52, Black: Bd5.** (a) Our position; Melanie plays Bd5 (e4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#994` (ply 93) [structural distance $d=3$, QAP obj=15]: the master played Bf6; under μ our *Be5* plays the role of that game’s *Bb7*; transferred move e4d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3159` (ply 96) [structural distance $d=5$, QAP obj=20]: the master played Bd8; under μ our *Be5* plays the role of that game’s *Bc7*; cross-type roles: our *Be4* as their *Pc2*; transferred move e4h1. (d) Retrieved from `ccr1_top_p0003#1726` (ply 96) [structural distance $d=2$, QAP obj=16]: the master played Bd5; under μ our *Be5* plays the role of that game’s *Be6*; transferred move e4f3. Synthesis: 1 retrieved analog transferred to Bd5 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



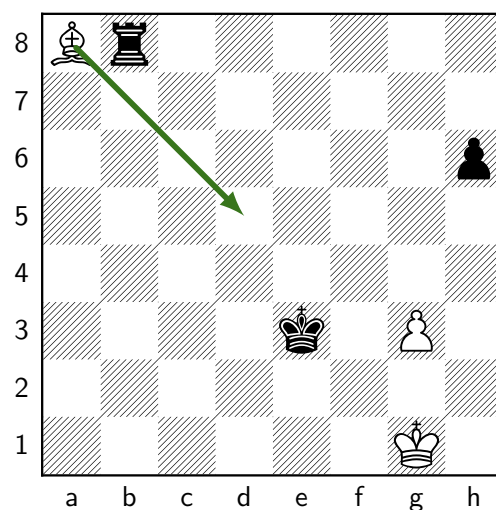
(a)



(b)

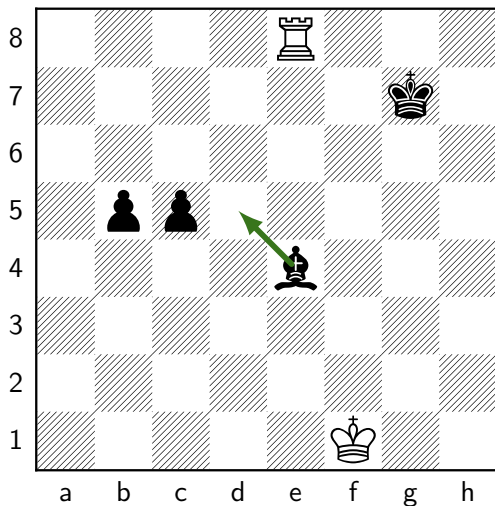


(c)

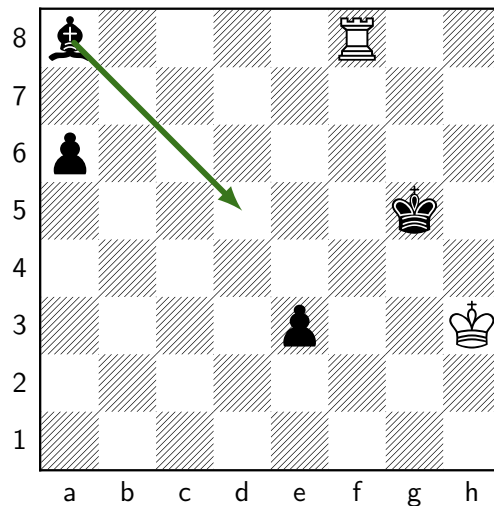


(d)

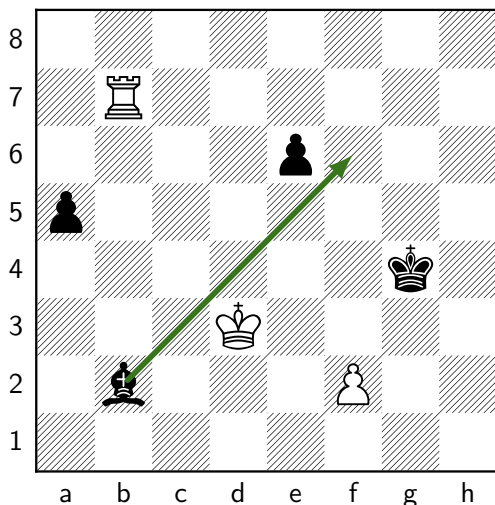
Figure 47: **Move 53, Black: Be4**. (a) Our position; Melanie plays **Be4** (**d5e4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3412` (ply 121) [structural distance $d=5$, QAP obj=24]: the master played **d4**; under μ our *Bd4* plays the role of that game’s *Pd4* (a **cross-type** correspondence); transferred move **d5e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1726` (ply 96) [structural distance $d=2$, QAP obj=13]: the master played **Bd5**; under μ our *Bd4* plays the role of that game’s *Be6*; transferred move **d5e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3664` (ply 106) [structural distance $d=3$, QAP obj=7]: the master played **Bd5**; under μ our *Bd4* plays the role of that game’s *Ba8*; transferred move **d5e4** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Be4** (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



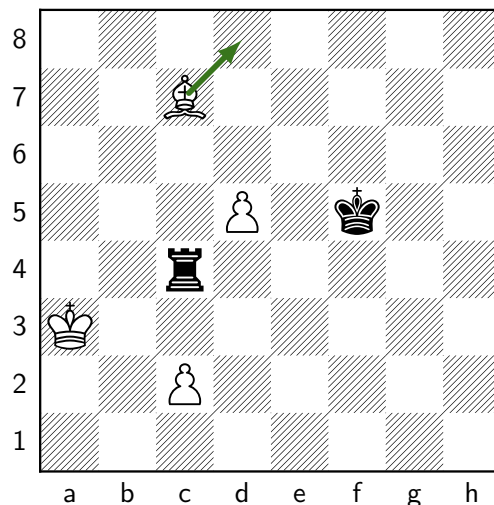
(a)



(b)



(c)



(d)

Figure 48: **Move 54, Black: Bd5.** (a) Our position; Melanie plays Bd5 (**e4d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1707` (ply 123) [structural distance $d=1$, QAP obj=16]: the master played Bd5; under μ our *Be5* plays the role of that game’s *Ba1*; transferred move **e4d5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#994` (ply 93) [structural distance $d=3$, QAP obj=15]: the master played Bf6; under μ our *Be5* plays the role of that game’s *Bb7*; transferred move **e4d5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3159` (ply 96) [structural distance $d=5$, QAP obj=20]: the master played Bd8; under μ our *Be5* plays the role of that game’s *Bc7*; cross-type roles: our *Be4* as their *Pc2*; transferred move **e4h1**. Synthesis: 2 retrieved analogs transferred to Bd5 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.